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PROJECT TITLE

Implementing a Multifaceted, Comprehensive, Appropriate Care Package for People with Advanced and Terminal illnesses needing ICU support and knowing its benefits on Quality of Dying, Family Satisfaction, Care Cost and Resource Utilisation.

RESEARCH TEAM

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Research Locations – Year 1 of the project

Kasturba Medical College, Manipal

St John's Medical College Hospital, Bangalore

DATE OF STARTING

15/09/2025

The study was registered in Clinical Trials Registry- India (CTRI) with registration number CTRI/2025/06/088645 [Registered on: 11/06/2025]

PROJECT DURATION

Total Project Duration: 60 months

Reporting Period Covered: 9 months.

OBJECTIVES

Objective 1: Co-develop the MCACP intervention through iterative, participatory design.

Duration

- 0-6 months

Objective 2: Pilot the MCACP intervention in two ICUs with existing palliative care integration to refine implementation and intervention, and explore the key facilitators and barriers.

Duration

6 months – 18 months, phased into:

- 0–3 months: Co-design adaptation
- 4–12 months: Pilot rollout and iterative learning
- 13–18 months: Consolidated refinements and readiness assessment

Objective 3: Expand implementation to four ICUs with existing palliative care access, with the aim of refining implementation and the intervention, and evaluating its effectiveness on quality of dying, family satisfaction, cost, and resource utilisation.

Duration

- 12 – 36 months (Including formative phase, implementation, and pre-post evaluation)

Objective 4: Extend implementation to four ICUs without existing palliative care access, with an aim of refining implementation and intervention, and evaluate its effectiveness on the quality of dying, family satisfaction, cost, and resource utilisation.

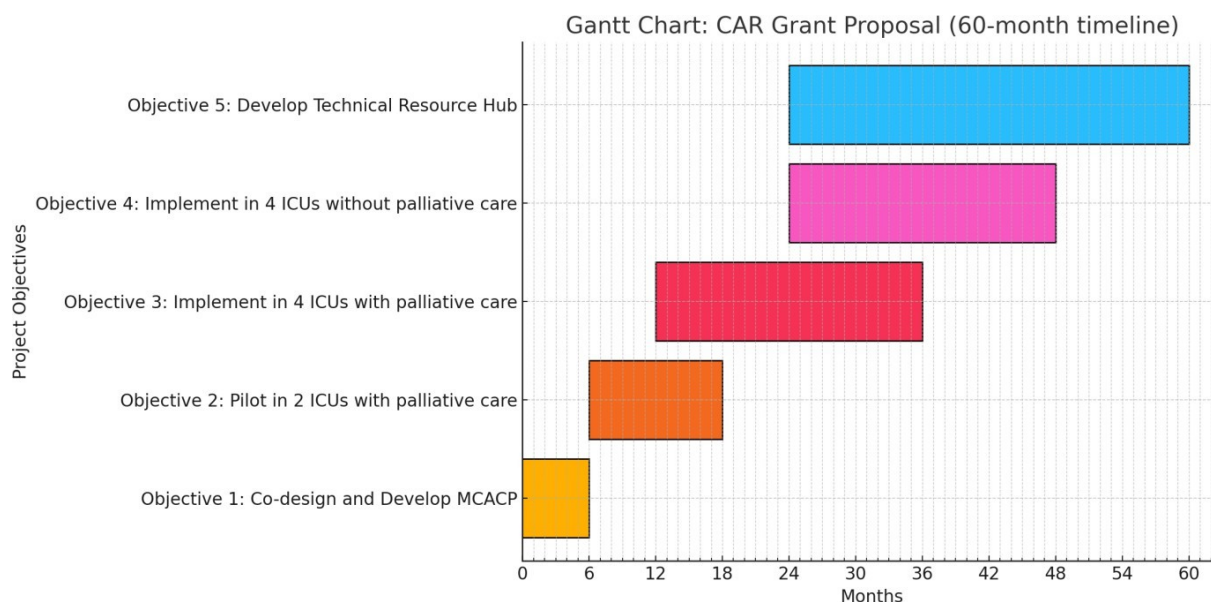
Duration

- 24 – 48 months

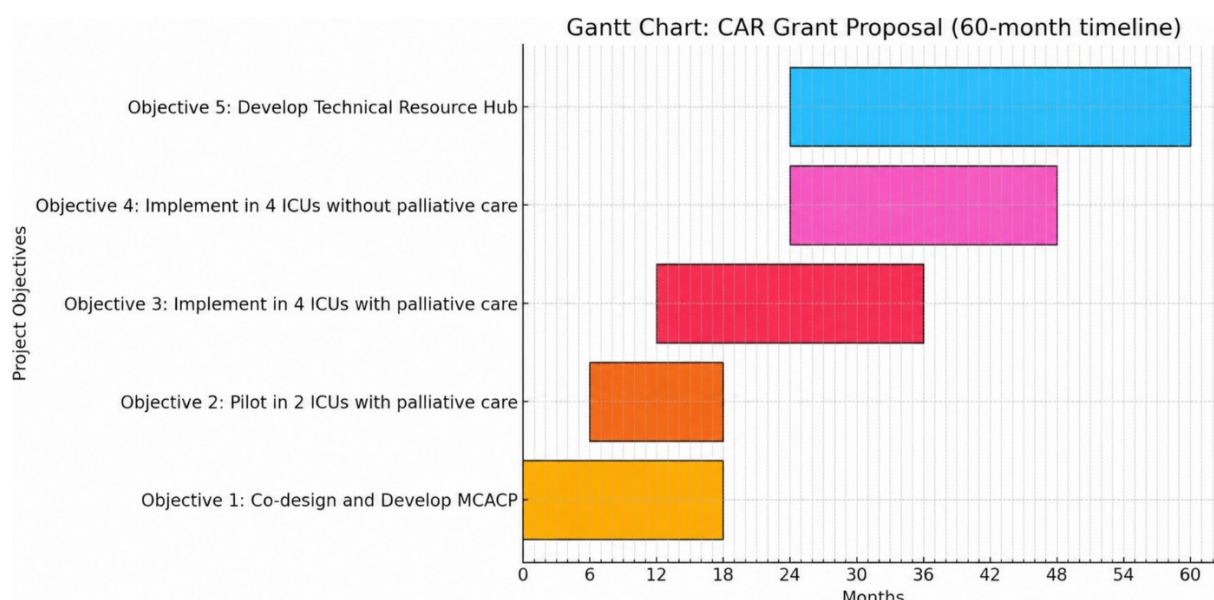
Objective 5: Develop a Technical Resource Hub for Supporting ICUs Across India

Duration

- 24 – 60 months



Gantt Chart – Modified – Objective 1 to 18 months - INTERIM MODIFICATION OF OBJECTIVES / METHODOLOGY



Work Package (WP)	Planned Activities	Current Status	Progress
WP1. Evidence synthesis and intervention development	Conduct six realist reviews; develop programme theory; develop draft MCACP intervention	Six realist reviews completed; preliminary programme theory developed; draft MCACP intervention package being prepared	Completed (80%)
WP2. Development of the MCACP manual and implementation toolkit	Develop SOPs, algorithms, consent forms, documentation templates, implementation workflows, and educational resources	SOPs, questionnaires, consent forms, documentation templates and workflows completed; MCACP manual under development and being refined using pilot findings	Substantially completed (~80–90%)
WP3. Stakeholder engagement and co-design	Multidisciplinary consultations, expert meetings, stakeholder workshops, iterative refinement	Multidisciplinary consultations completed; stakeholder engagement undertaken; implementation workshops initiated; continuous refinement ongoing	Substantially completed (~85%)
WP4. Site preparation and implementation readiness	Ethics approval, site preparation, staff training, implementation planning, workflow preparation	Ethics approvals completed; site initiation completed; implementation teams established; workflow preparation completed at both sites	Completed (100%)
WP5. Baseline mixed-methods assessment	Quantitative baseline data collection; qualitative interviews; organisational assessment	KMC: 43 quantitative participants recruited; SJMC: recruitment initiated (3 participants); qualitative interviews: KMC 23 completed,	Ongoing (~50–60%)

Work Package (WP) Planned Activities		Current Status	Progress
WP6. Pilot implementation of MCACP	Implement intervention in KMC and SJMC; workflow integration; implementation monitoring	SJMC 18 completed; organisational assessment ongoing	Ongoing (~45–50%)
		Pilot implementation initiated in both centres; workflow testing ongoing; implementation workshops underway; intervention embedded into routine practice is being evaluated	
WP7. Process evaluation and implementation research	Assess feasibility, acceptability, fidelity, adoption, barriers, facilitators, workflow integration	Qualitative interviews completed in both sites; key barriers and facilitators identified; workflow observations ongoing; implementation evaluation in progress	Ongoing (~50–60%)
WP8. Refinement of the MCACP intervention	Refine intervention using pilot findings and stakeholder feedback; finalise implementation package	Preliminary refinements identified (prognostication, communication, documentation, family meetings, referral triggers); refinement continues as additional pilot data are collected	Ongoing (~40–50%)
WP9. Preparation for multicentre implementation	Develop implementation package, readiness assessment, manuals, and training resources	Initial planning commenced; implementation package evolving alongside pilot refinement; multicentre network established	Initiated (~20–30%)

Publication of the overall protocol (status of the same)

Implementation of a Multifaceted, Comprehensive, and Suitable Care Package for Patients with Advanced and Terminal Illnesses in Intensive Care Units in India: A Multi-Phase Sequential Mixed-Methods Study Protocol

Implementation of a Multifaceted, Comprehensive, and Suitable Care Package for Patients with Advance

CURRENT STATUS

Your revised submission has passed our technical checks

Your revised submission has passed the technical checks and is with the editor that will review before making a decision or send it for peer review.

The length of time this takes can depend on several factors, including finding the best peer reviewers for your submission.

Progress so far

[Show history](#)

- Submission received
- Technical check
- Editorial assignment
- With editor

METHODOLOGY

Study Design:

Mixed-methods implementation research programme incorporating qualitative, quantitative, implementation science, and health economics approaches.

Frameworks Used:

- Consolidated Framework for Implementation Research (CFIR)
- Normalisation Process Theory (NPT)
- Realist approaches for understanding implementation mechanisms

Methods:

- Nominal Group Technique
- Qualitative interviews and focus group discussions
- Prospective observational study
- Quasi-experimental implementation studies
- Cost and resource-utilisation analyses
- Implementation fidelity and readiness assessments

DETAILED PROGRESS OF WORK CARRIED OUT DURING THE REPORTING PERIOD

Objectives 1 and 2: Months 0–18

Objective 1

Work Package 1: Evidence Synthesis and Intervention Development

The MCACP intervention was developed using the UK Medical Research Council (MRC) Framework for Complex Interventions. Six realist reviews were conducted to identify the contexts, mechanisms, and outcomes that influence the provision of appropriate end-of-life care in intensive care units. The reviews examined six core domains: prognostication, communication, shared decision-making, physician consensus, care planning, and care delivery. Realist synthesis was used to develop context–mechanism–outcome (CMO) configurations that explain how and why interventions improve appropriate care across different ICU settings. Findings from these reviews informed the development of the initial programme theory and the first draft of the Multifaceted Comprehensive Appropriate Care Package (MCACP).

Work Package 2: Development of the MCACP Manual and Implementation Toolkit

Evidence generated from the realist reviews is being synthesised into a structured implementation package comprising seven intervention components: prognostication, physician consensus, communication, shared decision-making, comprehensive anticipatory care planning, care delivery, and care evaluation. Standard operating procedures (SOPs), clinical algorithms, documentation templates, consent forms, case report forms, workflow pathways, implementation checklists, educational resources, and training materials are being developed to support standardised implementation. The manual is being designed as a living document that undergoes iterative refinement based on findings from pilot implementation. Forms are attached as a zip folder.

Work Package 3: Stakeholder Engagement and Co-design

A participatory co-design approach was adopted to ensure that the intervention was contextually appropriate and acceptable to end users. Multidisciplinary stakeholders, including intensivists, palliative care physicians, nurses, hospital administrators, implementation scientists, legal experts, patients' family representatives, and researchers, participated in consultations, expert meetings, and implementation workshops. Human-centred design principles and the Consolidated Framework for Implementation Research (CFIR) guided the identification of barriers, facilitators, and contextual adaptations required for successful implementation.

Work Package 4: Site Preparation and Pilot Implementation

Pilot implementation was initiated in two tertiary-care ICUs with established palliative care services: Kasturba Medical College (KMC), Manipal, and St. John's Medical College Hospital (SJMC), Bengaluru. Preparatory activities included obtaining ethics approvals, site initiation, staff orientation, establishment of implementation teams, workflow mapping, and development of local implementation plans. The intervention was integrated into routine ICU workflows using iterative Plan–Do–Study–Act (PDSA) cycles to optimise feasibility and minimise disruption to clinical practice. The field visit by the KMC-Manipal research team to St John's Hospital (21 April 2026) supported clarification of study processes and key challenges, monitoring of study progress, documentation, shared understanding of the project, and planning the objectives for the next 6 months.

Work Package 5: Baseline Mixed-Methods Evaluation

A convergent mixed-methods study was conducted to establish baseline clinical practices and implementation readiness. Quantitative data were collected on patient demographics, illness trajectories, ICU resource utilisation, symptom burden, palliative care integration, quality of dying, and family satisfaction using validated instruments. Qualitative data were obtained through semi-structured interviews with ICU physicians, nurses, palliative care clinicians, hospital administrators, patients' family members, and other stakeholders to explore experiences of end-of-life care, organisational readiness, barriers, facilitators, and implementation needs. Quantitative data were analysed using descriptive statistics, while qualitative interviews were analysed using reflexive thematic analysis informed by CFIR and realist evaluation principles.

Work Package 6: Process Evaluation and Implementation Research

Implementation was evaluated using a mixed-methods implementation science approach guided by CFIR, Normalisation Process Theory (NPT), and the Implementation Research Logic Model (IRLM). Data sources included implementation logs, workflow observations, meeting records,

fidelity assessments, staff feedback, and qualitative interviews. Process evaluation focused on implementation outcomes, including feasibility, acceptability, appropriateness, adoption, fidelity, usability, and workflow integration. Continuous feedback was used to identify barriers and facilitators and inform iterative refinement of intervention components.

Work Package 7: Intervention Refinement and Manual Development

Findings from the realist reviews, baseline assessments, pilot implementation, workflow observations, and stakeholder feedback are being synthesised to refine the MCACP intervention. Attention was given to optimising prognostication pathways, multidisciplinary physician consensus mechanisms, structured communication processes, shared decision-making tools, anticipatory care planning documentation, symptom management protocols, and quality evaluation indicators. The refined intervention was incorporated into successive versions of the MCACP manual together with implementation guidance, educational resources, audit indicators, and fidelity measures. This iterative refinement will continue until completion of the pilot phase and preparation for multicentre implementation.

Application of Implementation Research Frameworks in the Development of the MCACP, please see annexure 1

Realist reviews

Focus group discussions were conducted for each realist review – Doctors, Nurses, Allied Healthcare Professionals, Administrators, Caregivers with Lived Experience, and Technical Site Experts. Stakeholder engagement was coordinated by the CAR research team (CoPIs and research scientists) to contextualise findings from the realist reviews and to ensure that the MCACP reflected the realities of ICU practice in India. Topic guides were developed for each realist review based on the emerging Context–Mechanism–Outcome configurations and were refined following discussions among investigators.

Purposive sampling was used to recruit stakeholders representing the multidisciplinary ICU team, including intensivists, palliative care physicians, nursing officers, hospital administrators, social workers, psychologists (where available), and family caregivers of critically ill patients. Participants were selected to ensure representation from both open and closed ICU models and from centres with varying levels of palliative care integration (all ten clinical sites).

Semi-structured interviews and focus group discussions (FGDs) were conducted over MS Teams. Moderators had prior experience conducting qualitative research through the Breathe India programme and received additional orientation regarding realist interviewing techniques and end-of-life care research. An observer (project technical officer) attended each FGD to document group dynamics, non-verbal interactions, and contextual observations.

Each FGD followed a structured discussion guide exploring current end-of-life care practices, barriers to implementation, multidisciplinary decision-making, communication processes, organisational culture, legal concerns, documentation practices, and opportunities for improvement. Sessions typically lasted 60–90 minutes and were conducted via MS Teams to facilitate open discussion.

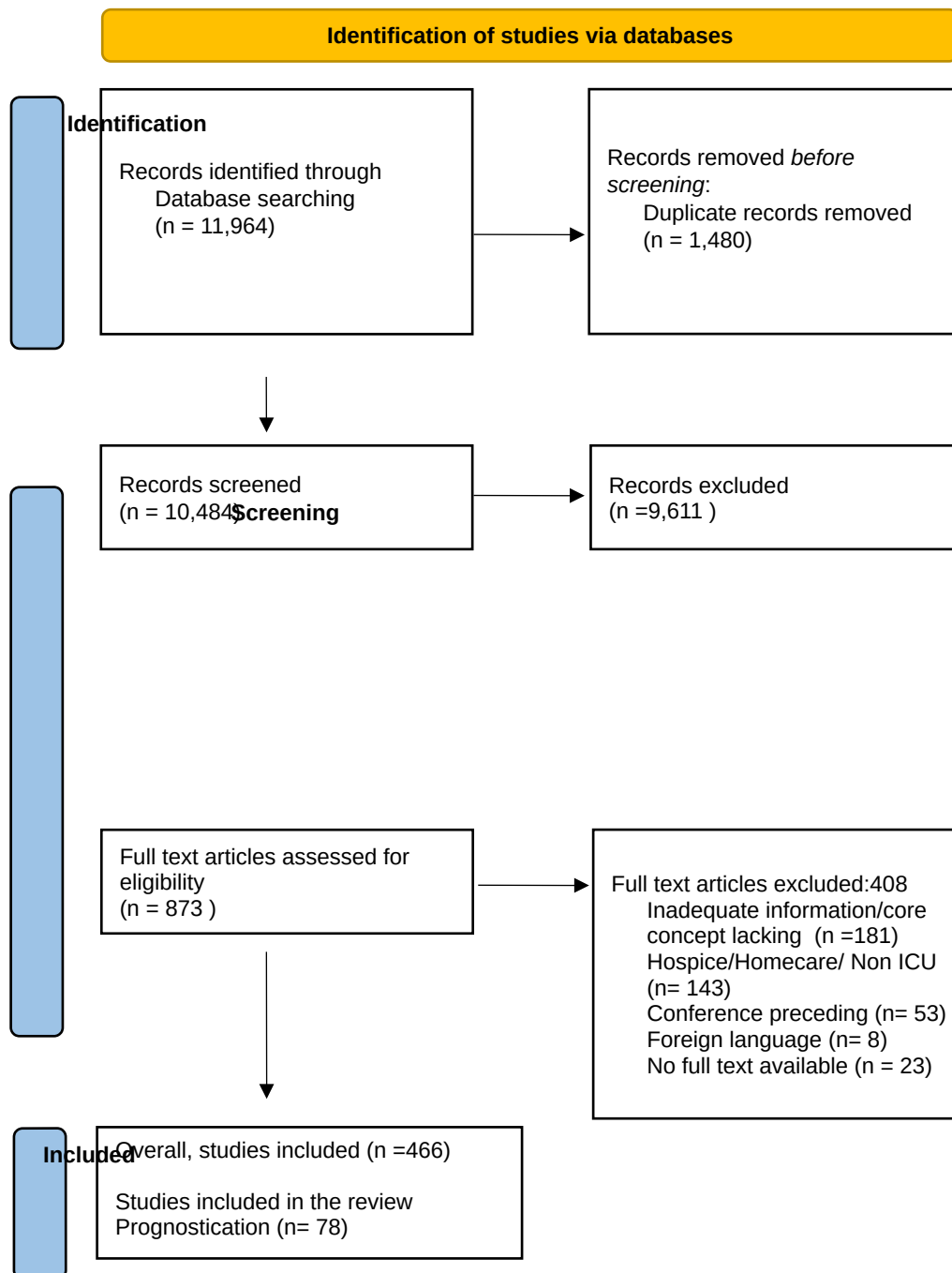
Items	Physicians (number participated)	Nurses (number participated)	Administrators and public health experts (number participated)	Allied health professionals, like physiotherapists, occupational therapists, and psychologists (number participated)	Caregivers/ members of the public with lived experience (number participated)	Key Output for MCACP
Prognostication	13	17	12	10	5	Prognostication methods and contextual factors
Communication	13	17	12	10	5	Communication guide
Shared Decision-Making	13	17	12	10	5	Shared decision-making toolkit
Physician Consensus	13	17	12	10	5	Consensus Care Review Board (CCRB) model
Care Planning	13	17	12	10	5	Comprehensive Advance Care Planning template
Care Delivery	13	17	12	10	5	Care delivery pathway, post-ICU care

Following consent, all interviews and FGDs were digitally recorded. Moderators (from the research team) maintained detailed field notes documenting participant interactions, implementation issues, emerging concepts, and contextual observations. Recordings were transcribed verbatim, anonymised, and verified for accuracy by the research team before analysis.

Qualitative data were managed using NVivo. Coding was undertaken independently by two researchers (from our team) using an inductive–deductive thematic approach informed by realist methodology and the Consolidated Framework for Implementation Research (CFIR). Regular coding meetings were conducted to resolve discrepancies and refine the coding framework. Emerging themes were mapped against the six realist reviews and subsequently synthesised to refine the MCACP intervention.

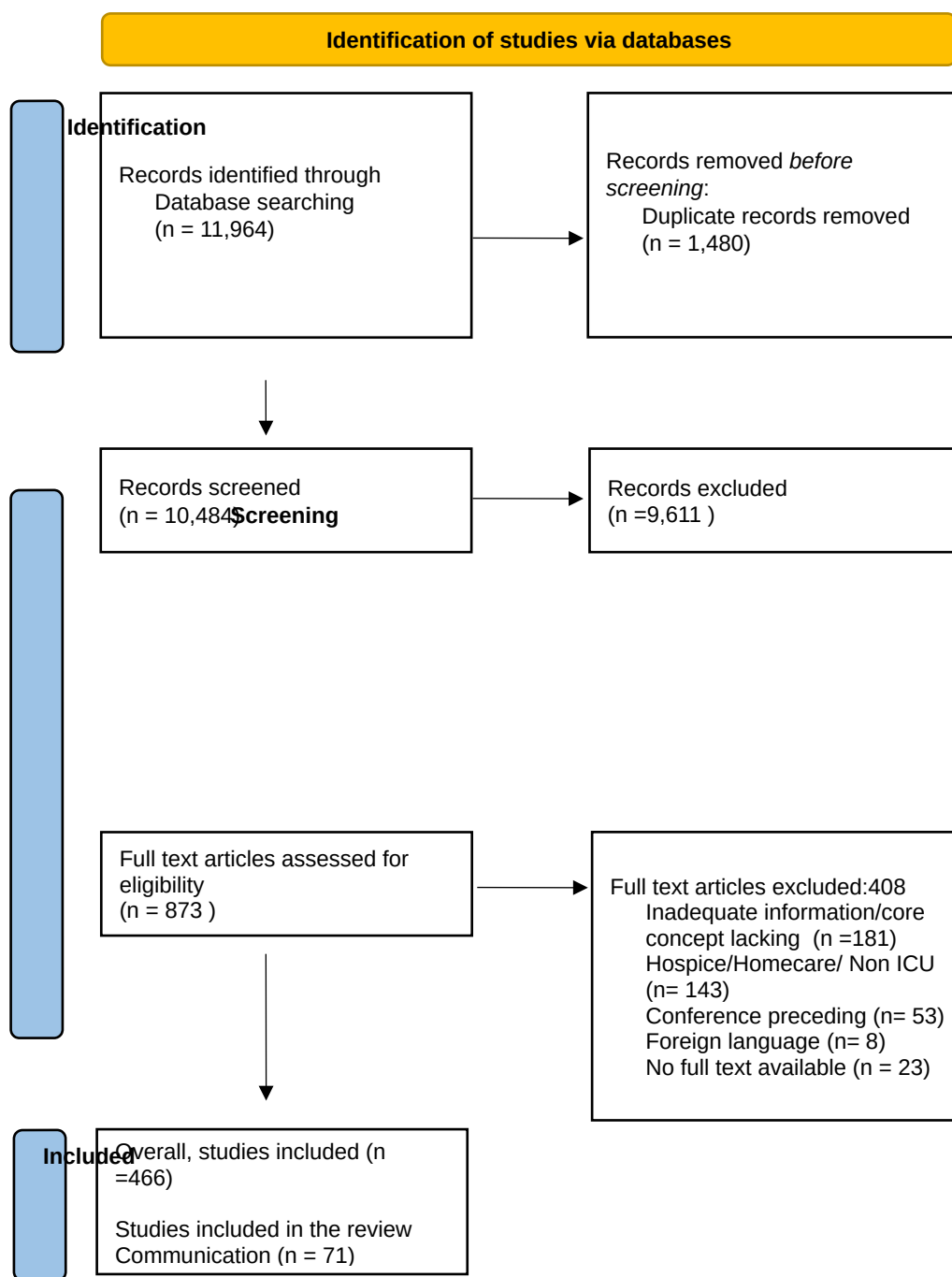
Preliminary findings from stakeholder engagement were presented at multidisciplinary investigator meetings (every Friday) that included intensivists, palliative care physicians, and research staff from KMC and SJMC. These meetings enabled iterative refinement of intervention components before pilot implementation. Stakeholder feedback will directly inform the development of prognostication triggers, multidisciplinary review processes, communication pathways, family meeting templates, documentation formats, symptom management algorithms, and implementation workflows, which will be included in the first draft of the MCACP manual.

PROGNOSTICATION



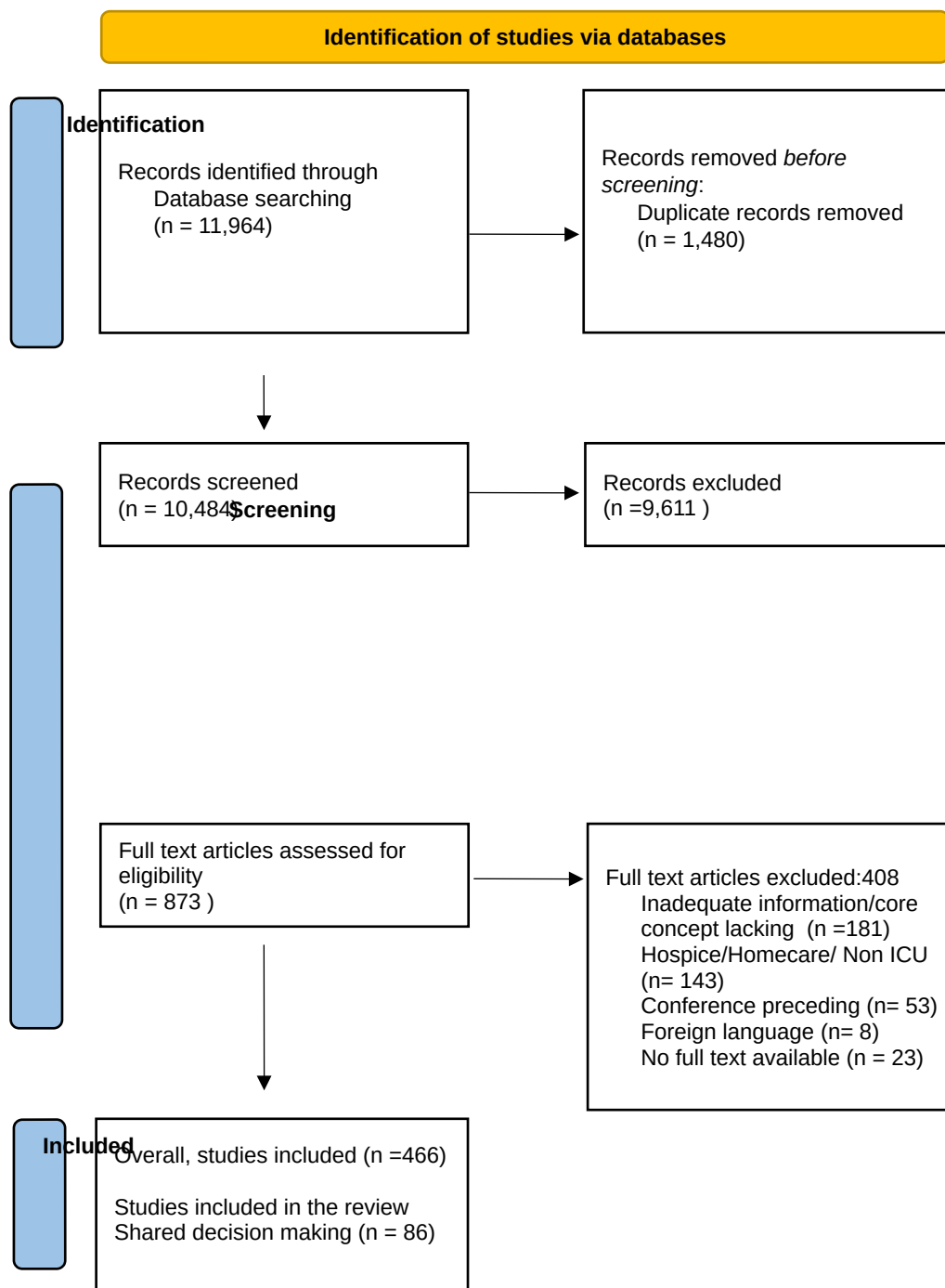
The prognostication realist review identified 78 studies that examined how prognostic assessment influences decision-making in critically ill patients. The evidence suggests that prognostication is a complex and dynamic process that combines clinical expertise, objective prediction tools, patient-specific factors, and interdisciplinary input. The review highlighted that uncertainty in prognosis often delays discussions of goals of care and the integration of palliative care, while structured prognostic approaches may facilitate timely and appropriate treatment decisions.

COMMUNICATION



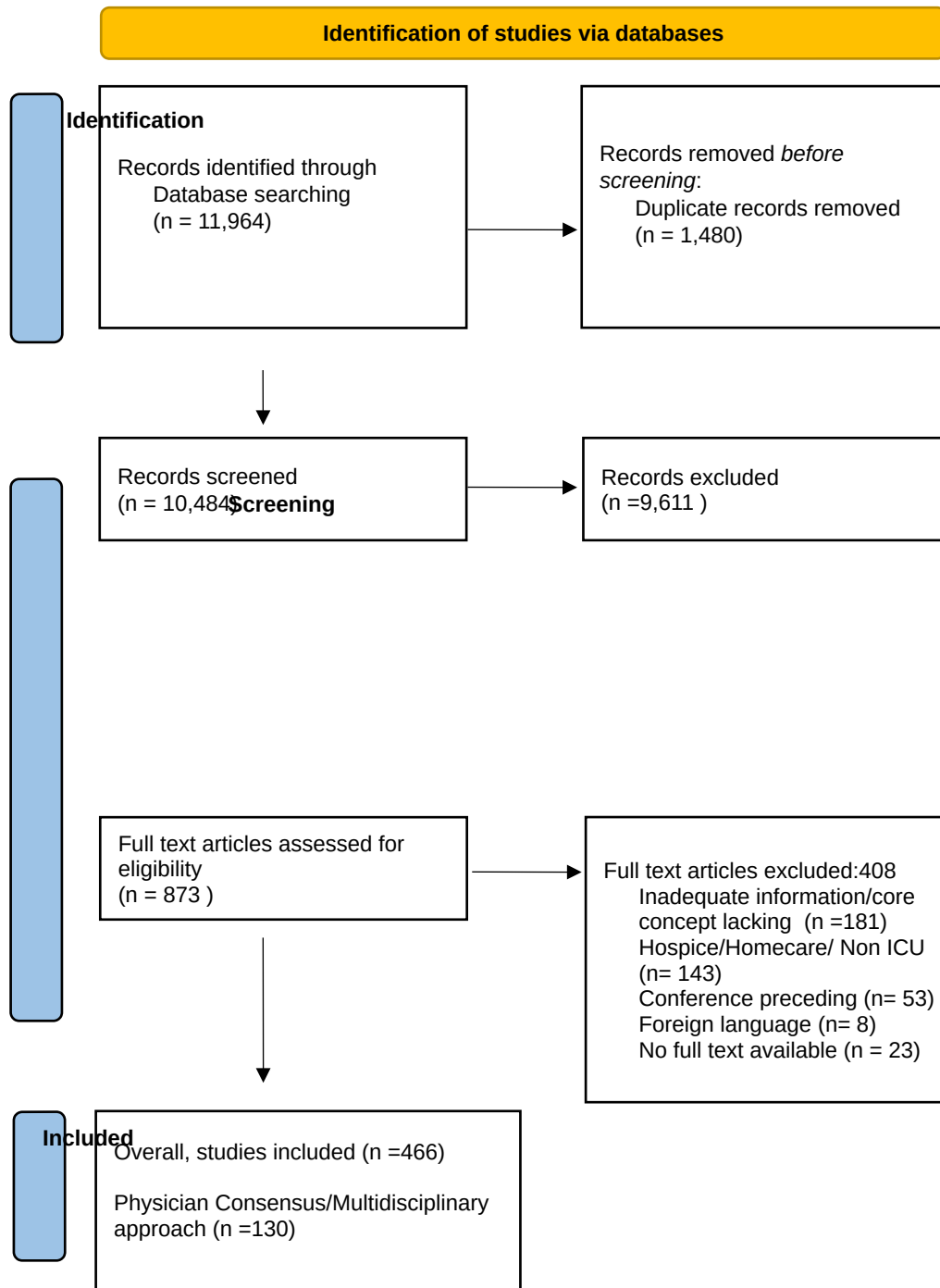
The communication realist review included 71 studies evaluating communication practices between healthcare professionals, patients, and families in the intensive care setting. Preliminary findings indicate that effective communication is central to establishing trust, aligning treatment with patient values, and reducing conflict during critical illness. Interventions such as structured family meetings, communication training, and standardized discussion frameworks were frequently associated with improved satisfaction and better shared understanding of treatment goals.

SHARED DECISION MAKING



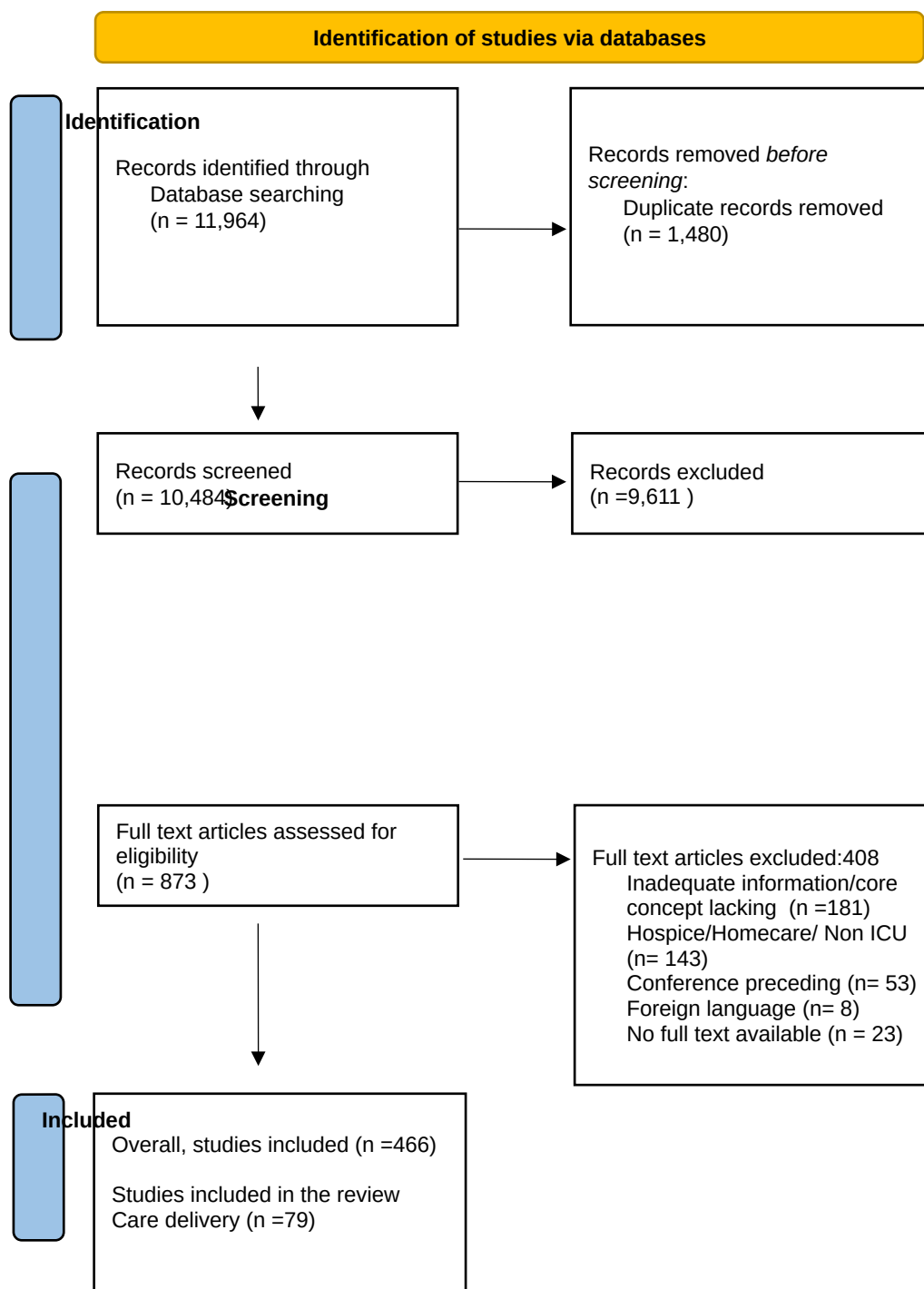
The shared decision-making realist review synthesized evidence from 86 studies examining how patients, families, and healthcare teams participate in treatment decisions. The findings suggest that successful shared decision-making depends on clear communication, accurate prognostic information, family engagement, and clinician willingness to explore patient preferences. Decision-making was found to be most effective when supported by multidisciplinary collaboration and when adequate time and resources were available for meaningful discussions.

PHYSICIAN CONSENSUS/MULTIDISCIPLINARY APPROACH



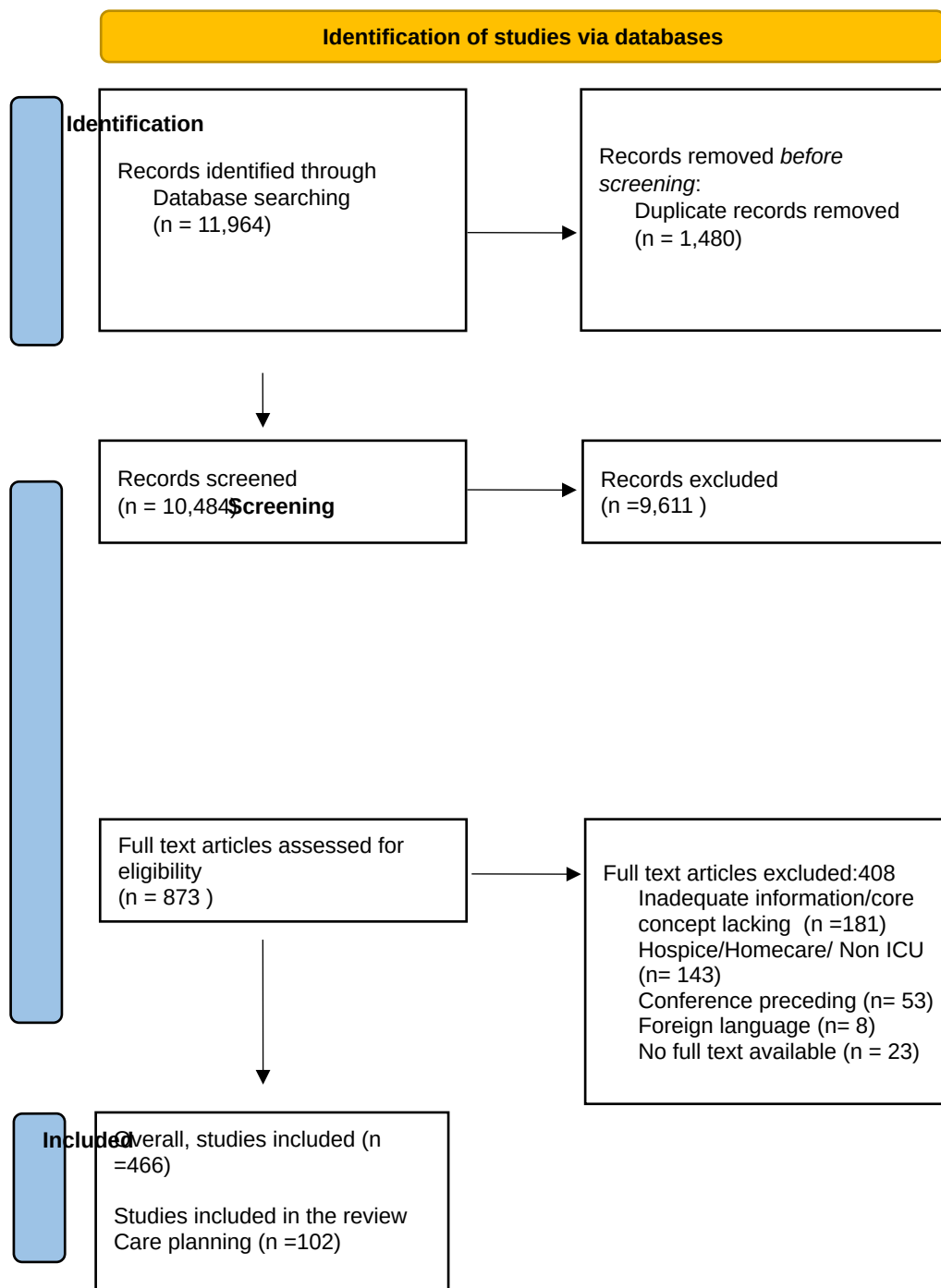
The physician consensus and multidisciplinary approach realist review represented the largest body of evidence, comprising 130 studies. Preliminary findings demonstrate that coordinated multidisciplinary decision-making, including contributions from intensivists, palliative care specialists, nurses, and other allied health professionals, promotes more consistent, patient-centred care. Regular multidisciplinary meetings and consensus-building processes were associated with improved treatment planning, reduced variability in clinical practice, and greater confidence in complex end-of-life decisions.

CARE DELIVERY



The care delivery realist review identified 79 studies evaluating models of integrating palliative care into intensive care practice. The evidence suggests that care delivery is strengthened through early identification of patients with palliative care needs, standardised referral pathways, multidisciplinary collaboration, and organisational support. Integrated models of care were consistently associated with improvements in symptom management, coordination of care, family support, and efficient utilisation of healthcare resources.

CARE PLANNING



The care planning realist review included 102 studies focusing on advance care planning, goals-of-care discussions, and individualised treatment planning. Preliminary findings indicate that proactive care planning facilitates treatment decisions that are better aligned with patients' values and preferences while reducing unnecessary interventions at the end of life. Successful care planning was supported by early conversations, repeated reassessment throughout the illness trajectory,

comprehensive documentation, and active involvement of patients, families, and multidisciplinary healthcare teams.

Realist Review	Studies Included (n)	Preliminary Result
Prognostication	78	Seventy-eight studies were identified that addressed prognostication in ICU and palliative care settings, providing evidence on prognostic tools, clinician judgement, and outcome prediction.
Communication	71	Seventy-one studies focused on communication between healthcare professionals, patients, and families, examining approaches to goals-of-care discussions and information sharing.
Shared Decision-Making	86	Eighty-six studies addressed shared decision-making, exploring patient and family involvement in treatment decisions, advance care planning, and collaborative decision processes.
Physician Consensus / Multidisciplinary Approach	130	This was the largest evidence base, with 130 studies examining multidisciplinary collaboration, consensus-building among clinicians, ethics consultation, and interdisciplinary ICU decision-making.
Care Delivery	79	Seventy-nine studies evaluated models of care delivery, including integration of palliative care into ICU practice, service organisation, and implementation strategies.
Care Planning	102	One hundred and two studies focused on care planning, including advance care planning, goals-of-care documentation, and individualised treatment planning.

Summary of Realist Review Preliminary Results

Six realist reviews were undertaken to examine the evidence underpinning appropriate end-of-life care in intensive care units, focusing on prognostication, communication, shared decision-making, physician consensus, comprehensive anticipatory care planning, and care delivery. A systematic search of multiple electronic databases identified 11,964 records, of which 466 studies met the inclusion criteria and were synthesised using realist methodology. Across the six reviews, several recurring themes emerged. Appropriate end-of-life care was consistently facilitated by early recognition of patients with poor prognosis, structured multidisciplinary communication, timely family meetings, shared decision-making that aligned treatment with patient values, and integration of specialist palliative care within routine ICU practice. Organisational factors, including leadership support, established institutional policies, availability of trained personnel, and standardised documentation processes, were identified as critical contextual determinants influencing implementation. Conversely, prognostic uncertainty, inconsistent communication, limited clinician confidence, inadequate documentation of goals of care, legal concerns regarding treatment

limitation, and variability in organisational workflows were recurrent barriers across healthcare settings. The reviews also highlighted that successful implementation depended not on individual interventions alone but on coordinated, multifaceted approaches addressing clinical, behavioural, organisational, and health-system factors simultaneously.

Roadmap to the Development of the MCACP Manual

The MCACP manual will be developed through an iterative, evidence-informed process that integrates findings from six realist reviews with implementation learning from two pilot ICUs (KMC Manipal and St. John's Medical College, Bengaluru). Rather than producing a static guideline, the manual will evolve through continuous refinement, ensuring that every component is supported by both international evidence and real-world implementation experience in Indian ICUs.

Step 1: Building the Evidence Base through Realist Reviews

The six realist reviews provide the theoretical foundation for the MCACP by identifying what interventions work, for whom, under what circumstances, and why. The prognostication review informs the development of standardised criteria for recognising patients approaching the end of life and establishing referral triggers. The communication review contributes structured approaches for family meetings, difficult conversations, conflict resolution, and culturally appropriate communication. The shared decision-making review informs family engagement strategies, decision aids, and documentation processes. The physician consensus review provides evidence to establish multidisciplinary Critical Care Review Boards (CCRBs), physician consensus pathways, and ethical deliberation processes. The care planning review contributes to the framework for Comprehensive Anticipatory Care Planning (CACP), including documentation of treatment limitations, ceilings of care, advance care planning, and consent processes. Finally, the care delivery review provides evidence for symptom management protocols, bereavement care, psychosocial support, and integration of specialist palliative care into routine ICU workflows.

Step 2: Contextualisation through Pilot Implementation

The pilot implementation in KMC Manipal and St. John's Medical College tests the feasibility and contextual suitability of these evidence-based components within Indian ICU settings. Quantitative data evaluate patient characteristics, resource utilisation, symptom burden, family satisfaction, quality of dying, and implementation outcomes, while qualitative interviews with clinicians, patients' families, administrators, and stakeholders identify practical barriers and facilitators to implementation. Simultaneously, workflow observations examine how each intervention can be incorporated into routine ICU practice without disrupting existing clinical care.

Step 3: Refining Individual MCACP Components

The preliminary pilot findings demonstrate important gaps that directly inform refinement of the manual. Although almost all patients required intensive life-sustaining treatments, specialist palliative care consultations (11.6%), documented prognosis discussions (18.6%), formal family meetings (9.3%), and advance directives were uncommon, indicating the need for clearer referral triggers, structured communication pathways, and standardised documentation tools. High symptom burden—including pain (53.5%), respiratory secretions (37.2%), and agitation (32.6%)—supports development of comprehensive symptom assessment and management protocols. While families generally reported positive experiences regarding professionalism and communication, opportunities remain to improve comfort, dignity, emotional support, and quality of dying, reinforcing the need for anticipatory care planning, bereavement support, and family-centred care pathways.

Step 4: Developing the MCACP Manual

Evidence from the realist reviews and pilot implementation will be synthesised to produce the MCACP manual, consisting of seven integrated implementation modules: (1) Prognostication; (2) Physician Review and Consensus; (3) Communication; (4) Shared Decision-Making; (5) Comprehensive Anticipatory Care Planning; (6) Care Delivery; and (7) Care Evaluation. Each module will include standard operating procedures, clinical algorithms, decision aids, communication guides, documentation templates, educational resources, implementation checklists, training videos, audit indicators, and fidelity measures. The manual will distinguish between core components that should be implemented across all ICUs and adaptable components that can be modified according to local resources and organisational readiness.

Step 5: Usability Testing and Iterative Adaptation

Draft versions of the manual will undergo iterative usability testing with ICU physicians, nurses, palliative care specialists, administrators, and family representatives using human-centred design principles. Feedback from frontline users regarding clarity, feasibility, workload, acceptability, and workflow integration will be incorporated through repeated refinement cycles. This process will ensure that the MCACP manual remains practical, context-sensitive, and adaptable across diverse ICU environments, including open and closed ICU models and hospitals with differing levels of palliative care integration.

Step 6: Preparing for National Scale-up

The final MCACP manual will serve as the implementation package for subsequent multicentre implementation across ICUs with and without existing palliative care services. It will also form the foundation of the national Technical Resource Hub, supporting standardised training, implementation, quality improvement, and ongoing adaptation across India. Thus, the manual will represent the culmination of a sequential evidence-generation pathway, moving from realist evidence synthesis, through pilot implementation and iterative refinement, to a nationally scalable implementation resource to support the equitable integration of palliative care into intensive care practice.

PUBLICATIONS ARISING FROM THE PROJECT

Published

Ghoshal A, Prasad A, Nanda A, *et al* How prognostic information influences care planning in adult intensive care units: protocol for a realist review. *BMJ Open* 2026;**16**:e117449. doi: 10.1136/bmjopen-2026-117449

Publication history

Received January 27, 2026

Accepted May 29, 2026

First published June 29, 2026.

Online issue publication June 29, 2026

in review

A Realist Review Protocol of Communication Practices in Intensive Care Units

STATUS	ID	TITLE	CREATED	SUBMITTED
✉ Contact Journal EPA: Editorial Office, BMJ Open	bmjopen-2026-123169	A Realist Review Protocol of Communication Practices in Intensive Care Units View Submission [View Original Files] Cover Letter	25-May-2026	28-May-2026
Awaiting PRM Assignment				

Advance Care Planning in Intensive Care Units for People with Serious Life-Limiting Illnesses: A Realist Scoping Review Protocol

✉ Contact Journal EPA: Editorial Office, BMJ Open	bmjopen-2026-123179	Advance Care Planning in Intensive Care Units for People with Serious Life-Limiting Illnesses: A Realist Scoping Review Protocol View Submission [View Original Files] Cover Letter	25-May-2026	25-May-2026
Awaiting PRM Assignment				

Protocol: Realist Review of Shared Decision Making in Intensive Care Units

✉ Contact Journal EPA: Editorial Office, BMJ Open	bmjopen-2026-120758	Protocol: Realist Review of Shared Decision Making in Intensive Care Units View Submission [View Original Files] Cover Letter	06-Apr-2026	06-Apr-2026
Awaiting ME Decision				

Structured ICU Governance Mechanisms to Strengthen End-of-Life Decision-Making: A Realist Review Protocol

✉ Contact Journal EPA: Editorial Office, BMJ Open	bmjopen-2026-119520	Structured ICU Governance Mechanisms to Strengthen End-of-Life Decision-Making: A Realist Review Protocol View Submission [View Original Files] Cover Letter	12-Mar-2026	12-Mar-2026
Awaiting Reviewer Reply				

Objective 2: Pilot the MCACP Intervention in Two ICUs with Existing Palliative Care Integration

Work Package 2.1: Baseline Evaluation and Contextual Assessment

Methodology

A mixed-methods baseline assessment is being undertaken in two tertiary ICUs with established palliative care services (KMC Manipal and St. John's Medical College, Bengaluru). The objective is to understand existing ICU practices related to appropriate care, identify contextual determinants of implementation, and establish baseline patient, family, clinician, and organisational outcomes. The

study follows the UK Medical Research Council framework for complex interventions and is guided by the Consolidated Framework for Implementation Research (CFIR) and Normalisation Process Theory (NPT).

Participant Recruitment

Patients were screened daily by trained research staff in collaboration with ICU clinicians.

Eligible participants included adult ICU patients meeting predefined criteria for advanced life-limiting illness or those considered at high risk of mortality based on clinical judgement and ICU admission characteristics. Recruitment was undertaken using consecutive sampling to minimise selection bias.

Recruitment Progress till June 15, 2026

Site	Patients Screened	Participants Recruited
KMC Manipal	117	43
SJMC Bengaluru	25	15
Total	142	58

Data Collection

Dedicated research coordinators employed through the CAR programme conducted daily data collection under the supervision of the site principal investigators.

Clinical information was obtained through:

- electronic medical records
- bedside patient assessment
- ICU observation charts
- nursing documentation
- laboratory records
- medication charts
- family discussions where appropriate

Standardised Case Report Forms (CRFs) were developed specifically for the project and piloted prior to commencement.

Variables collected included:

Patient Characteristics

- age
- gender
- diagnosis
- illness trajectory
- comorbidities
- frailty
- reason for ICU admission

ICU Care

- mechanical ventilation
- vasopressor support
- renal replacement therapy
- invasive procedures
- ICU length of stay

Palliative Care Indicators

- specialist palliative care referral
- prognosis discussions
- family meetings
- treatment limitation decisions
- advance directives
- goals-of-care documentation

Symptom Assessment

- pain
- dyspnoea
- agitation
- respiratory secretions
- delirium

Family Outcomes

- family satisfaction
- communication quality
- quality of dying

Daily monitoring continued until ICU discharge or death.

Qualitative Study

Study Design

A qualitative descriptive study using semi-structured interviews was undertaken to understand the experiences, perceptions, and implementation needs of key stakeholders involved in ICU end-of-life care.

The qualitative work was embedded within the implementation study and informed iterative refinement of the MCACP.

Participant Sampling

Purposive sampling was used to ensure representation from all stakeholder groups involved in ICU decision-making.

Participants included:

- ICU consultants
- critical care fellows
- resident doctors
- ICU nurses
- palliative care physicians
- hospital administrators
- family caregivers
- implementation team members

Sampling continued until thematic saturation was achieved within stakeholder groups.

Recruitment Progress for qualitative interviews

Stakeholder Group	KMC	SJMC
ICU Consultants	6	3
ICU Nurses	5	4

Stakeholder Group	KMC	SJMC
Palliative Care Physicians	2	2
Hospital Administrators	4	2
Family Caregivers	5	9
Social Workers/others	1	4
Total Interviews	23	24

Data Collection

Semi-structured interview guides were developed using findings from the six realist reviews and aligned with the Consolidated Framework for Implementation Research (CFIR).

Separate interview guides were prepared for clinicians, nurses, administrators, and family caregivers.

Interviews explored:

- current ICU practices
- communication experiences
- prognostication
- shared decision-making
- treatment limitation
- legal concerns
- organisational culture
- workflow barriers
- multidisciplinary collaboration
- acceptability of MCACP
- recommendations for improvement

Interviews were conducted by trained qualitative researchers employed under the CAR programme. Prior to commencement, interviewers underwent training in qualitative interviewing, reflexivity, ethical research conduct, and realist interviewing techniques.

Each interview lasted approximately 45–75 minutes.

All interviews were:

- audio-recorded
- transcribed verbatim
- anonymised
- verified for accuracy
- stored in password-protected databases

Field notes documenting contextual observations, interviewer reflections, and implementation issues were maintained after every interview.

Qualitative Data Analysis

Qualitative analysis followed an iterative thematic analysis approach.

The analysis consisted of:

Step 1

Repeated reading of transcripts for familiarisation.

Step 2

Line-by-line open coding.

Step 3

Development of the initial codebook.

Step 4

Independent coding by two researchers.

Step 5

Consensus meetings to resolve coding differences.

Step 6

Grouping of related codes into categories.

Step 7

Generation of overarching themes.

Coding was informed by both inductive thematic analysis and deductive implementation science constructs derived from CFIR and the realist programme theory.

Example Coding Framework

Participant Quote	Initial Code	Category	Theme
"We usually wait until families ask..."	Delayed prognosis discussion	Communication practices	Delayed goals-of-care discussions
"Different consultants often have different opinions."	Variable physician opinions	Decision-making	Lack of multidisciplinary consensus
"Documentation is different for every consultant."	Inconsistent documentation	Organisational workflow	Lack of standardisation
"Families expect us to continue everything."	Unrealistic expectations	Family engagement	Communication and shared decision-making

Preliminary Themes

Thematic analysis identified several recurring themes:

- Prognostic uncertainty delays initiation of appropriate care.
- Communication with families remains largely unstructured.
- Decisions are influenced by multiple clinicians with limited formal consensus mechanisms.
- Documentation of goals of care is inconsistent.
- Specialist palliative care involvement occurs late in the illness trajectory.
- Clinicians expressed a strong desire for structured communication pathways and practical implementation guidance.
- Existing ICU workflows require intervention components that are simple, flexible, and integrated into routine practice.

Integration of Quantitative and Qualitative Findings

Quantitative and qualitative findings were reviewed jointly during multidisciplinary investigator meetings.

The quantitative data demonstrated low rates of discussions of prognosis, family meetings, advance care planning, and palliative care referrals despite high symptom burden.

The qualitative interviews explained these findings by identifying organisational barriers, prognostic uncertainty, workflow limitations, documentation challenges, legal concerns, and variability in multidisciplinary decision-making.

This triangulation directly informed refinement of the MCACP intervention, particularly the development of prognostication triggers, communication guides, multidisciplinary physician review pathways, documentation templates, and implementation workflows.

Planned Publications

Baseline ICU palliative care practices and implementation readiness in two Indian tertiary ICUs: A mixed-methods study.

Work Package 2.2: Pilot Implementation and Workflow Optimisation

Methodology

The preliminary MCACP intervention is being implemented using an iterative co-development approach. Implementation focuses on integrating the seven MCACP components into existing ICU workflows while identifying operational challenges, implementation barriers, and opportunities for adaptation. Continuous learning cycles involving frontline clinicians, nurses, implementation champions, and palliative care specialists are undertaken to optimise delivery across both open and closed ICU models.

Data Collection and Analysis

Implementation outcomes include intervention fidelity, acceptability, feasibility, adoption, usability, workflow integration, staff engagement, and training completion. Data sources include implementation logs, structured observations, field notes, workflow mapping, staff feedback, and qualitative interviews. Qualitative findings are analysed thematically using CFIR and NPT constructs to understand mechanisms that facilitate or hinder successful implementation. Field observations: To understand the ICU setting, we observed the ICU workflow, admission process, billing process, counselling, bedside nursing care, transitions, and medical board meetings. A framework to record observational field notes was prepared for this.

Preliminary Results

Implementation activities are ongoing at KMC Manipal, with workflow testing and refinement of intervention components. At St. John's Medical College, implementation workshops have commenced following site initiation. Early observations indicate the need for simplified documentation, clearly defined referral triggers, structured multidisciplinary family meetings, and workflow integration strategies that minimise disruption to routine ICU practice while improving consistency of appropriate care delivery.

Planned Publications

Implementation of the Multifaceted Comprehensive Appropriate Care Package (MCACP): Early implementation experiences from two pilot ICUs in India.

Work Package 2.3: Refinement of the MCACP Intervention and Manual Development

Methodology

Findings from the six realist reviews, baseline assessments, implementation experience, workflow observations, and stakeholder feedback will be synthesised to iteratively refine the MCACP intervention. Human-centred design methods, Delphi consensus, and stakeholder workshops will be used to ensure the intervention remains context-sensitive, feasible, and scalable. Refinements will be documented using the TIDieR framework to facilitate reproducibility and future implementation.

Data Collection and Analysis

Quantitative implementation metrics and qualitative stakeholder feedback will be triangulated to identify intervention components requiring modification. Consensus meetings will determine the core intervention elements, optional context-specific adaptations, implementation tools, educational resources, and quality indicators to be incorporated into the final MCACP manual.

Preliminary Results

The pilot work has confirmed several priority areas requiring refinement, including structured prognostication pathways, multidisciplinary physician consensus mechanisms, standardised communication guides, shared decision-making tools, comprehensive anticipatory care planning templates, symptom management algorithms, and care evaluation indicators. These findings are directly informing the development of the first draft of the MCACP manual and associated implementation toolkit.

Planned Publications

1. *Development of the MCACP intervention manual using realist evidence and pilot implementation findings.*
2. *Co-design and refinement of a comprehensive ICU palliative care intervention for India: A mixed-methods implementation study.*

Publications -

Vallath N, Parvathy V, Padhi P, Johny T A, Krishnakumar M, Varghese PS. Procedures to implement the Supreme Court of India directives to withdraw/withhold life-sustaining medical treatment. *Indian J Med Res.* 2026 Mar;163(3):319-326. doi: 10.25259/IJMR_3005_2025. PMID: 42024905; PMCID: PMC13126532.

Quality Improvement project: addressing quality gaps in the documentation of medical certificates of cause of death in the palliative medicine department, IP setting, to improve patient care: led by Dr Angeline (DNB Palliative medicine) and mentored by Dr Nandini Vallath. Drafting of the manuscript is ongoing.

A **Quality Improvement (QI) study** was undertaken to reduce the **Discharge Against Medical Advice (DAMA)** rate through collaboration between the **Department of Pain and Palliative Medicine and Supportive Care**, the **Critical Care Department**, and the **Surgical and Neuro Critical Care Departments**. The study identified key drivers contributing to DAMA and implemented strategies to address them. The findings and key drivers from this QI initiative will be incorporated into the ongoing ICMR study to inform interventions to reduce DAMA rates and improve continuity of care for critically ill patients transitioning to palliative care (under preparation for publication).

Presentation in conference

- Dr. Amarja Havaladar, presented paper on Quality improvement study to reduce incidence of discharge against medical advice (DAMA) from intensive care units of a tertiary academic centre, using the A3 methodological approach in *Criticare 2026*
- Dr. Angelin Johny, presented paper on Determinants of discharge against medical advice (DAMA) in ICU: A prospective quality improvement study on modifiable contributors in *33rd IAPCON 2026*.

Project Personnel and Responsibilities During the Reporting Period

Kasturba Medical College (KMC), Manipal (12 Personnel)

1. Site Implementation Lead Support (1)

The Site Implementation Lead Support coordinated the day-to-day implementation of the CAR project at KMC. Responsibilities included coordinating study activities across the ICU and palliative care teams, monitoring adherence to study timelines, organising investigator meetings, supervising recruitment activities, resolving operational issues, facilitating implementation of the MCACP intervention, and ensuring communication between the principal investigators and research staff. The role also involved supporting workflow integration, documenting implementation challenges, and coordinating iterative refinement of intervention components.

2. Training Programme Coordinator (1)

The Training Programme Coordinator was responsible for planning, organising, and delivering educational activities related to the MCACP intervention. Responsibilities included preparing training schedules, developing educational materials, organising orientation programmes for clinicians and research staff, maintaining attendance records, coordinating simulation sessions where applicable, evaluating training outcomes, and ensuring the competency of staff involved in intervention delivery.

3. Data Management and Reporting Coordinator (1)

The Data Management and Reporting Coordinator established and maintained the project database and ensured data quality throughout the study. Responsibilities included database development, preparation of electronic case report forms, data validation, quality assurance checks, data cleaning, generation of recruitment reports, preparation of progress reports, maintenance of study documentation, and supporting interim analyses. The coordinator also monitored data completeness across study sites and prepared reports for investigators, ethics committees, and the funding agency.

4. Qualitative Research Coordinator (1)

The Qualitative Research Coordinator led all qualitative components of the study. Responsibilities included developing interview guides, scheduling interviews and focus group discussions, training interviewers, supervising qualitative data collection, managing transcription, quality-assuring transcripts, supervising coding, conducting thematic analysis, maintaining qualitative databases, and preparing qualitative summaries to inform intervention refinement.

5. Digital Content Liaison (1)

The Digital Content Liaison developed and managed educational and communication resources required for implementation. Responsibilities included preparing digital learning materials, managing training videos, updating online educational resources, coordinating dissemination materials, maintaining project communication platforms, supporting webinar organisation, and assisting with digital documentation of project activities.

6. Liaison for Multi-site Communication (1)

This role ensured effective coordination between participating institutions. Responsibilities included scheduling regular multicentre meetings, maintaining communication between investigators, tracking project milestones across sites, coordinating document sharing, preparing meeting minutes,

monitoring action points, facilitating collaborative decision-making, and supporting harmonisation of study procedures across centres.

7. Regulatory and Ethics Documentation Officer (1)

The Regulatory and Ethics Documentation Officer coordinated all regulatory activities. Responsibilities included preparing and submitting ethics applications, maintaining regulatory files, tracking approvals and renewals, ensuring compliance with institutional and national regulations, preparing protocol amendments, maintaining investigator site files, documenting adverse events where applicable, and ensuring adherence to Good Clinical Practice (GCP) documentation standards.

8. Project Financial and Administrative Support (1)

The Project Financial and Administrative Support Officer managed the administrative aspects of the CAR grant. Responsibilities included financial documentation, budget monitoring, procurement of study materials, processing reimbursements, maintaining expenditure records, coordinating travel and logistics for project meetings, supporting human resource documentation, and ensuring compliance with institutional financial procedures and ICMR reporting requirements.

9. Project Research Scientist – Implementation (1)

The Project Research Scientist (Implementation) coordinated the implementation of the MCACP intervention at the clinical level. Responsibilities included supporting participant recruitment, supervising intervention delivery, monitoring implementation fidelity, conducting workflow observations, facilitating multidisciplinary meetings, documenting implementation barriers and facilitators, supporting clinician engagement, and contributing to refinement of intervention components based on implementation experience.

10. Project Research Scientist – Evaluation (1)

The Project Research Scientist (Evaluation) was responsible for monitoring study outcomes and evaluating implementation processes. Responsibilities included oversight of quantitative data collection, quality assurance of clinical data, statistical analyses, implementation outcome evaluation (feasibility, acceptability, fidelity, adoption), integration of quantitative and qualitative findings, preparation of reports, and contribution to manuscript development.

11. Medical Social Worker (MSW) (1)

The Medical Social Worker supported participant engagement and the project's family-centred aspects. Responsibilities included identifying eligible family participants, coordinating informed consent, facilitating qualitative interviews with caregivers, supporting family meetings where appropriate, documenting psychosocial concerns, assisting with follow-up assessments, and acting as a liaison between research staff and families during data collection.

12. Office Helper (1)

The Office Helper provided logistical and administrative support essential for the smooth conduct of project activities. Responsibilities included organising study materials, maintaining project records, supporting meeting logistics, assisting with document management, facilitating the

movement of study documents between departments, coordinating office supplies, and supporting day-to-day administrative operations.

St. John's Medical College (SJMC), Bengaluru (4 Personnel)

1. Project Research Scientist – Implementation (1)

The Project Research Scientist coordinated the implementation of the MCACP intervention at SJMC. Responsibilities included participant screening and recruitment, supervision of intervention delivery, coordination with ICU clinicians, workflow mapping, implementation monitoring, documentation of implementation challenges, and participation in regular multicentre review meetings.

2. Project Research Scientist – Evaluation (1)

The Project Research Scientist (Evaluation) oversaw quantitative and implementation evaluation activities. Responsibilities included data quality monitoring, participant follow-up, implementation outcome assessment, statistical analyses, preparation of interim reports, and integration of evaluation findings into intervention refinement.

3. Research Psychologist (1)

The Research Psychologist led the psychosocial and qualitative components at SJMC. Responsibilities included conducting semi-structured interviews with family caregivers and healthcare professionals; assessing communication experiences; exploring psychological and emotional aspects of end-of-life care; facilitating qualitative data collection; assisting with coding and thematic analysis; and contributing to the development of the MCACP's communication and family support components.

4. Research Nurse (1)

The Research Nurse supported clinical implementation and participant recruitment within the ICU. Responsibilities included daily screening of eligible patients, collection of clinical and nursing data, maintenance of study documentation, coordination of informed consent procedures, monitoring adherence to the study protocol, assisting with symptom assessments, facilitating communication between the clinical and research teams, and ensuring completeness of bedside data collection.

EXECUTIVE SUMMARY

Title

Implementing a Multifaceted, Comprehensive, Appropriate Care Package for People with Advanced and Terminal Illnesses Needing ICU Support and Knowing its Benefits on Quality of Dying, Family Satisfaction, Care Cost and Resource Utilisation.

Introduction

Patients with advanced and terminal illnesses frequently receive intensive and potentially non-beneficial interventions in ICUs. Integration of palliative care within ICU practice remains inconsistent in India, contributing to increased suffering, financial burden, and inefficient resource utilisation.

Rationale

The project aims to develop and evaluate a structured, Multifaceted, Comprehensive, Appropriate Care Package (MCACP) that integrates palliative care principles into ICU practice to improve patient, family, and health-system outcomes.

Objectives

To co-develop, pilot, refine, implement, and evaluate an ICU palliative care intervention package across multiple Indian centres and create a national implementation resource hub.

Methodology

Mixed-methods implementation science programme involving intervention co-design, pilot implementation, qualitative and quantitative evaluation, health economics, and multi-centre expansion using CFIR and NPT frameworks.

Results

- MCACP intervention developed and is being refined.
- 58 patients recruited, and 47 qualitative interviews completed (together at two sites).
- Significant gaps identified in palliative care integration, prognosis discussions, advance care planning, and family meetings.
- Preliminary findings demonstrate substantial unmet palliative care needs in ICU settings.
- Multi-centre implementation infrastructure has been established and is being implemented to bridge this gap.

Website is online/ being updated regularly <https://mathru.manipal.edu/>

- Translational Potential

The project is developing a scalable, contextually adaptable model for integrating palliative care in the ICU, implementable across tertiary, secondary, and resource-constrained healthcare settings in India. Findings are expected to inform national policy, clinical practice guidelines, training programmes, and healthcare delivery models.

RESEARCH WORK REMAINING TO BE COMPLETED

- Completion of pilot implementation at KMC and SJMC.
- Consolidated refinement of the MCACP intervention is ongoing.
- Expansion to additional participating centres.
- Completion of qualitative and quantitative recruitment.
- Implementation fidelity assessments.
- Health economics evaluation is planned.
- Development of a national technical resource hub.
- Multi-centre effectiveness evaluation.
- Knowledge translation and policy engagement activities.

APPLIED VALUE OF THE PROJECT

1. Education – ISCCM (Domain-specific PC in ICU Modules – provide module structure) EOLC & Palliative Care ICU Course (EPIC) – please see Annexure 2
2. Workshops at SJMC – qualitative research workshop, Implementation Science 101: Palliative care in Intensive Care Settings – please see Annexure 3

3. Legal framework – with Vidhi Centre for Legal Policy

In the intensive care unit (ICU), EOLC frequently involves complex decisions regarding withholding or withdrawing life-sustaining treatment, symptom management, communication with families, shared decision-making, and anticipatory care planning. In India, these decisions have traditionally been influenced not only by clinical uncertainty and cultural beliefs but also by concerns regarding legal liability, resulting in considerable variation in clinical practice and delayed integration of palliative care. We are working to simplify the process for healthcare workers by developing a Toolkit that provides hospitals with practical guidance on Advance Medical Directives, withholding and withdrawing life support, documentation, substitute decision-making, conflict resolution, and institutional governance.

23 April 2026: A training session was conducted for the Paediatric ICU and Paediatric Haematology-Oncology Department at SJMC to enhance understanding of the legal and institutional processes involved in withholding and withdrawal of life-sustaining treatment.

8 May 2026: As part of the 25th Anniversary of the Department of Forensic Medicine, a session on the Primary Medical Board (PMB) and Secondary Medical Board (SMB) process was conducted at SJMC. Dr Nandini Vallath delivered a lecture on the withholding and withdrawal of life-sustaining medical treatment, in accordance with the Supreme Court of India guidelines, for advocacy and awareness.

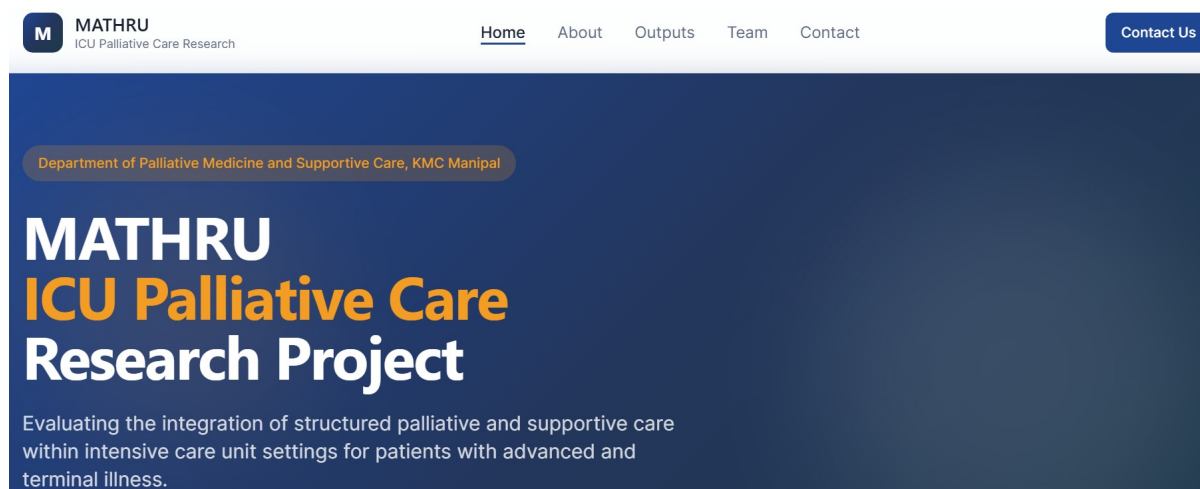
Living Will Document: A Living Will document was developed and published in the Indian Journal of Medical Research to facilitate implementation of the Supreme Court of India's directives on the withholding and withdrawal of life-sustaining medical treatment. The document serves as a practical resource to help individuals and family members record their preferences for future medical care if they are unable to communicate their decisions, thereby supporting advance care planning and informed decision-making.

Karnataka Association of Critical Care: Dr Nandini Vallath conducted a Focus Group Discussion (FGD) on the Primary Medical Board (PMB) and Secondary Medical Board (SMB) processes, addressing the legal and ethical aspects of withholding and withdrawing life-sustaining medical treatment. The session was held at the CG Hall, St. John's Medical College.

18 June 2026: An educational session was conducted at the St. John's Geriatric Centre for Medical Social Workers (MSWs) and caregivers of patients from St. John's Hospital and the St. John's Geriatric Centre. The session focused on Advance Care Planning, with particular emphasis on the Living Will, its purpose, legal framework, and its role in enabling individuals to document their future healthcare preferences.

4. ApTCare-ICU (Appropriate Transitions in Care for ICU Patients) tool, Framework for field notes, two additional forms were developed for patients who recovered for many months after ICU admission (St John's) - on going

5. Project website - <https://mathru.manipal.edu/>



6. Spin-off projects

One of the major achievements of the CAR programme has been the development of several postgraduate research projects that have emerged directly from the intervention development and pilot implementation phases. These projects address specific knowledge gaps identified during the realist reviews, stakeholder consultations, and pilot implementation, while contributing additional evidence to strengthen the Multifaceted Comprehensive Appropriate Care Package (MCACP). Collectively, these studies expand the scope of the CAR by generating complementary evidence on governance, education, communication, implementation science, and organisational readiness. The projects also contribute to capacity building by training future clinician-researchers in implementation science, critical care, and palliative care research. SJMC is working to expand the scope of the project by following up with patients who undergo treatment after transfer from the ICU to a Palliative care setting.

Spin-off Project	Origin within the CAR Programme	Contribution to the CAR
Critical Care Review Board (CCRB) Governance Model Study (PhD)	The physician consensus realist review and stakeholder consultations identified significant variability in multidisciplinary decision-making and a lack of structured governance mechanisms for complex end-of-life decisions. This led to the development of a doctoral project evaluating governance models for multidisciplinary Critical Care Review Boards.	The study will develop evidence-based governance structures, operational policies, accountability frameworks, and decision-making processes that will strengthen the physician consensus component of the MCACP and support wider implementation across Indian ICUs.
ICU Nursing Education in Palliative Care (PhD)	The communication and care delivery reviews highlighted the	Findings will inform the development of competency-

Spin-off Project	Origin within the CAR Programme	Contribution to the CAR
Integration of Specialist Palliative Care within Intensive Care (PhD)	central role of nurses in symptom assessment, family communication, and continuity of care, while identifying important educational gaps in palliative care competencies.	based educational modules, training resources, and implementation strategies for ICU nurses, strengthening workforce capacity within the MCACP.
	The pilot implementation demonstrated limited referrals to specialist palliative care despite a high burden of unmet palliative care needs. This highlighted the need to understand organisational models for integrating specialist palliative care into routine ICU practice.	The study will identify effective referral pathways, collaborative care models, and organisational strategies to improve integration of palliative care services, directly informing the care delivery component of the MCACP.
	The communication realist review and qualitative interviews identified communication as a dynamic, relational process influenced by contextual, cultural, and emotional factors. This prompted a focused qualitative study exploring clinician–family interactions in ICU settings.	Findings will refine communication frameworks, family meeting guides, and shared decision-making strategies within the MCACP by providing a deeper understanding of the interactional processes that occur during serious-illness conversations.
ICU Communication as a Dynamic Field: A Gestalt Analysis of Clinician–Family Interactions (PhD)	During site preparation, considerable variation was observed in organisational readiness, leadership support, staffing, workflows, and institutional culture across participating ICUs. This led to a focused implementation science project evaluating readiness for change.	The study will identify organisational determinants of successful implementation and contribute to the development of readiness assessment tools, implementation checklists, and site-preparation strategies for the future multicentre rollout of the MCACP.
Assessment of ICU Readiness for Implementing a Multifaceted Palliative Care Intervention: A Mixed-Methods Study (MD Thesis)	Evaluating the Acceptability, Feasibility, and Fidelity of Early Implementation of the MCACP Communication and Prognostication Bundle in the ICU: A Prospective Observational Study (MD Thesis)	Early pilot implementation identified the communication and prognostication bundle as the first components to be integrated into routine ICU practice. Evaluating these components separately provides an opportunity to optimise implementation before wider scale-up.

Spin-off Project	Origin within the CAR Programme	Contribution to the CAR programme.
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PATENTS APPLIED FOR

Nil

ADDITIONAL BUDGET / STAFF REQUIREMENT

No additional budget or staff support is currently requested. Project staff joined in January of 2026 for KMC and in February 2026 for SJMC. We have requested that ICMR reallocate funds within certain categories, as expenses vary over time and do not always align with our initial plan, while keeping the overall budget unchanged. As some staff members joined later than expected, we plan to request a no-cost extension to ensure their activities are completed and the project is finalised successfully. Any future needs arising from expansion to additional sites will be communicated to ICMR through proper channels, with detailed justification.

Date: _____

Signature of Principal Investigator

Dr. Naveen Salins

Professor & Head, Department of Palliative Medicine and Supportive Care, Kasturba Medical College, Manipal

1. Intervention Development

Framework: UK Medical Research Council (MRC) Framework for Complex Interventions

The development of the Multifaceted Comprehensive Appropriate Care Package (MCACP) followed the Medical Research Council (MRC) Framework for the development and evaluation of complex interventions.

Completed activities

Development phase

- Conducted six realist reviews to identify the mechanisms, contexts, and outcomes influencing appropriate end-of-life care in ICUs.
- Synthesised evidence into seven preliminary MCACP components:
 - o Prognostication
 - o Physician consensus
 - o Communication
 - o Shared decision-making
 - o Comprehensive anticipatory care planning
 - o Care delivery
 - o Care evaluation
- Developed an initial programme theory describing how each intervention component is expected to improve appropriate care within different ICU contexts.

The initial programme theory proposes that the Multifaceted Comprehensive Appropriate Care Package (MCACP) will improve the quality, consistency, and appropriateness of end-of-life care in Indian intensive care units by addressing the clinical, organisational, behavioural, legal, and communication factors that influence decision-making. The intervention recognises that ICU end-of-life care is a complex adaptive process involving multiple stakeholders operating within diverse organisational and cultural contexts. The programme theory assumes that when evidence-based clinical processes are integrated with multidisciplinary collaboration, structured communication, and implementation support, clinicians will be better able to deliver care that aligns with patients' goals, improves family experiences, and reduces non-beneficial interventions.

Component 1. Prognostication

Context

ICU clinicians often experience uncertainty when identifying patients approaching the end of life, particularly in the presence of multiple comorbidities, organ failure, and unpredictable illness trajectories. This uncertainty frequently delays palliative care referral and goals-of-care discussions.

Mechanism

Providing structured prognostic assessment tools, clinical triggers, and multidisciplinary review increases clinicians' confidence in recognising limited prognosis, facilitates earlier identification of patients requiring appropriate care, and supports timely initiation of advance care planning.

Expected Outcomes

- Earlier recognition of patients with a limited prognosis.
- Timely palliative care referral.
- Earlier goals-of-care discussions.

- Reduction in potentially non-beneficial intensive treatments.

Component 2. Physician Consensus

Context

End-of-life decisions in ICUs are frequently influenced by differences in opinion among treating specialists, intensivists, and consulting teams, resulting in inconsistent treatment recommendations and delayed decision-making.

Mechanism

Structured multidisciplinary physician review and consensus processes improve shared understanding of prognosis, reduce professional uncertainty, distribute decision-making responsibility, and enhance confidence in treatment limitation decisions.

Expected Outcomes

- Increased multidisciplinary agreement.
- Reduced variation in clinical decisions.
- Timelier treatment limitation decisions.
- Improved consistency of patient-centred care.

Component 3. Communication

Context

Communication regarding prognosis and treatment goals is often delayed, inconsistent, or unstructured, leading to misunderstanding, family conflict, and unrealistic expectations.

Mechanism

Structured communication guides, scheduled family meetings, and communication training improve clinicians' ability to explain prognosis, address uncertainty, explore patient values, and build trusting relationships with families.

Expected Outcomes

- Improved quality of family communication.
- Greater family understanding of prognosis.
- Reduced decisional conflict.
- Increased family satisfaction with ICU care.

Component 4. Shared Decision-Making

Context

Treatment decisions frequently remain clinician-driven because patients are unable to participate, family members have limited understanding of illness, or there is no structured decision-making process.

Mechanism

Decision aids, structured family engagement, and shared decision-making frameworks enable clinicians and families to jointly evaluate treatment options in light of patients' preferences, values, and goals of care.

Expected Outcomes

- Increased family participation in decision-making.
- Greater concordance between treatments and patient preferences.
- Reduced moral distress among clinicians and families.
- Improved quality of end-of-life decisions.

Component 5. Comprehensive Anticipatory Care Planning (CACP)

Context

Advance care planning and documentation of treatment preferences are uncommon in Indian ICUs, resulting in repeated crisis-based decision-making and inconsistent care.

Mechanism

Standardised anticipatory care planning templates and documentation processes facilitate proactive discussions about goals of care, treatment ceilings, advance directives, and future clinical decisions.

Expected Outcomes

- Increased documentation of care preferences.
- Earlier care planning discussions.
- Improved continuity across treating teams.
- Greater delivery of preference-concordant care.

Component 6. Care Delivery

Context

Patients approaching the end of life often experience high symptom burden, fragmented palliative care involvement, and variable standards of supportive care.

Mechanism

Standardised symptom management protocols, early integration of specialist palliative care, multidisciplinary care coordination, and psychosocial support improve the quality and consistency of patient-centred care.

Expected Outcomes

- Improved symptom control.
- Earlier palliative care involvement.
- Improved patient comfort and dignity.
- Reduced unnecessary invasive interventions.
- Better support for families during the dying process.

Component 7. Care Evaluation

Context

Most ICUs lack routine systems to evaluate the quality of end-of-life care, limiting opportunities for continuous quality improvement.

Mechanism

Routine measurement of implementation fidelity, quality-of-dying indicators, family satisfaction, communication quality, and resource utilisation enables clinicians and organisations to monitor performance, identify gaps, and continuously refine practice.

Expected Outcomes

- Improved adherence to MCACP.
- Higher quality-of-dying scores.
- Greater family satisfaction.
- Continuous quality improvement.
- Sustainable integration of appropriate care into routine ICU practice.

Feasibility phase (ongoing)

- Pilot implementation initiated in two tertiary ICUs with existing palliative care integration.
- Mixed-methods evaluation undertaken to determine feasibility, acceptability, usability, workflow integration, and implementation fidelity.
- Iterative refinement of intervention components based on stakeholder feedback and implementation experience.

The project is therefore progressing from the Development to the Feasibility phase of the MRC framework.

2. Context Assessment

Framework: Consolidated Framework for Implementation Research (CFIR)

CFIR has been used to understand contextual factors influencing implementation across the two pilot ICUs.

Intervention Characteristics

- Assessed the adaptability of MCACP to both open (KMC) and closed (SJMC) ICU models.
- Evaluated perceived complexity of intervention components.
- Identified opportunities to simplify documentation and workflows.

Outer Setting

- Explored patient and family expectations regarding end-of-life care.
- Identified legal and ethical influences affecting treatment limitation decisions.
- Examined the role of institutional policies and national end-of-life care guidance.

Inner Setting

- Assessed organisational culture regarding palliative care integration.
- Explored leadership engagement and multidisciplinary collaboration.
- Evaluated existing communication practices and ICU workflows.

Characteristics of Individuals

- Interviewed ICU physicians, nurses, palliative care clinicians, and administrators regarding:
 - o knowledge
 - o beliefs
 - o confidence
 - o perceived roles
 - o willingness to implement MCACP

Implementation Process

- Established local implementation teams.
- Initiated workflow testing.
- Conducted implementation workshops.
- Used continuous feedback to refine intervention delivery.

3. Co-design and Pilot Refinement

Framework: Human-Centred Design (HCD) + CFIR

The MCACP is being refined using a human-centred design approach that actively involves frontline users throughout implementation.

Completed activities include:

- Stakeholder interviews across both ICUs.
- Continuous workflow observation.

- Identification of implementation barriers and facilitators.
- Iterative refinement of:
 - o referral triggers
 - o communication pathways
 - o documentation templates
 - o multidisciplinary review processes
 - o symptom management workflows

Pilot findings have demonstrated the need for:

- structured family meeting templates
- earlier prognostication
- standardised documentation
- simplified care planning tools
- improved integration of specialist palliative care.

The intervention continues to undergo iterative refinement based on user experience.

4. Behavioural Implementation

Framework: Normalization Process Theory (NPT)

NPT has been used to understand how MCACP becomes embedded into routine ICU practice.

Coherence

Clinicians generally recognise the importance of improving end-of-life care and understand the purpose of MCACP.

Cognitive Participation

ICU physicians, nurses, palliative care teams, and administrators have participated in implementation workshops and qualitative interviews, demonstrating early engagement with the intervention.

Collective Action

Workflow testing is evaluating how MCACP components integrate into daily ICU practice, including multidisciplinary meetings, prognostication, communication, documentation, and symptom management.

Reflexive Monitoring

Continuous feedback from implementation teams is informing ongoing refinement of the intervention, ensuring that it remains practical, acceptable, and contextually appropriate.

5. Implementation Planning

Framework: Implementation Research Logic Model (IRLM)

Implementation planning has followed the IRLM by explicitly linking determinants, strategies, mechanisms, implementation outcomes, and clinical outcomes.

Determinants

Identified determinants include:

- limited structured prognostication
- infrequent palliative care referral
- inconsistent documentation
- absence of formal family meetings
- organisational workflow constraints

- clinician uncertainty regarding end-of-life decision-making

Implementation Strategies

Strategies currently being tested include:

- multidisciplinary implementation workshops
- workflow redesign
- nurse/clinician education
- structured communication tools
- standardised documentation
- trying to identify/assign implementation champions
- continuous audit and feedback

Mechanisms

These strategies are expected to improve:

- nurse/ clinician knowledge
- confidence
- multidisciplinary collaboration
- shared decision-making
- communication consistency
- organisational readiness

Implementation Outcomes

Current evaluation includes:

- feasibility
- acceptability
- fidelity
- usability
- workflow integration
- stakeholder engagement

Clinical Outcomes

Outcomes being monitored include:

- quality of dying
- family satisfaction
- symptom control
- palliative care utilisation
- treatment limitation practices
- resource utilisation

6. Implementation Strategies

Framework: ERIC (Expert Recommendations for Implementing Change)

Several ERIC implementation strategies have already been incorporated into the pilot implementation.

ERIC Strategy	Current Application
Conduct educational meetings	Implementation workshops were conducted at ICUs
Develop educational materials	Preliminary MCACP protocols and training resources being developed
Identify and prepare	Local ICU and palliative care implementation leads are being

ERIC Strategy	Current Application
champions	identified
Conduct local needs assessment	Baseline quantitative and qualitative assessments ongoing
Audit and provide feedback	Ongoing monitoring of recruitment, workflow, and implementation progress
Tailor strategies	Intervention adapted for open and closed ICU models
Facilitation	Research team providing continuous implementation support
Build stakeholder relationships	Active engagement of physicians, nurses, administrators, and families
Capture and share local knowledge	Qualitative interviews informing intervention refinement
Revise professional roles	Strengthening multidisciplinary collaboration and physician consensus pathways

Annexure 2

Background

Appropriate EOLC and palliative care for patients coming to the ICU are now established practices worldwide. This helps avoid inappropriate life-sustaining treatments in the ICU and mitigates suffering for patients and their families.

In India, there has been considerable progress over the last 20 years in developing and refining EOLC practice guidelines through professional societies such as ISCCM and IAPC. Patient advocacy and advocacy by these societies have helped frame landmark legal judgments that lend vital support to these guidelines. Despite this, uptake of these guidelines has been patchy and clinical practice at the ground level has not changed much.

ICMR has allocated funds to establish a Centre for Advanced Research (CAR), primarily tasked with developing and implementing a Multifaceted Comprehensive Appropriate Care Package (MACACP) nationwide.

As part of this project and in collaboration with ISCCM, an EOLC & Palliative Care ICU Course (EPIC) is planned.

Aim

To create and implement an EOLC & Palliative Care ICU Course (EPIC) that is focused on training intensivists in India.

Objectives

1. Create a competency-based curriculum for EPIC based on the best available evidence that is in consonance with current practice and legal requirements
2. Prepare a teaching program that is interactive and has an online and offline component. The online component will have 12 modules (each with 2 sessions) spread over 3-4 months. The offline component will be conducted over 2 days and will have a strong focus on clinical ethical reasoning and communication skills

Ses sio n	Sr. No .	Modules	D at es	TOPIC
1	1	What is Palliative Care? Why Does it Matter? Why palliative care in ICU?		Define palliative care (PC)
				Principles of PC – what, why, when, how
				Evaluate myths and assumptions about PC
				Present Rationale and Data for integrating PC in the ICU
				Describe when and how to utilise a subspecialty palliative care team (models for ICU-PC Integration)
	2	What is end-of-life care		Define EOLC (various definitions)
				Identifying objectives of good EOLC
				Principles and components of a good death
				Identifying palliative care needs in the ICU – Screening and Trigger Criteria for PC in the ICU

				Rapid Palliative care assessment in the ICU
				Nutrition/Hydration at EOL
2	3	Illness Trajectories and Prognostication		Explore the five trajectories of Dying
				Identifying palliative care needs through the illness trajectories
				Prognostication in chronic illnesses Disease-specific prognostic tools Performance status measures Recognising futility
	4	Ethical & Legal Issues in EOLC		Legal policies at the end of life care Policies and guidelines for withdrawing and withholding life-sustaining treatment Physician-assisted suicide Euthanasia
				Ethical issues in end of life care
				Evaluating capacity and identifying decision-makers in the ICU
3	5	Basic communication		Learn methods for communicating prognosis Six steps of the SPIKES model for BBN SOLER/NURSE model
	6	Complex Communication Situations		Advance care plan/advance directives
				Withdrawing and withholding life support
				DNR/DNI
				Anger and Conflict Management
4	7	Goals of care discussion and decision-making		Denial and Collusion
				Understand ways to elicit goals of care
				Eliciting and resolving concerns at the end of life care decision making (Consensus building)
	8	Handling difficult families		Steps in shared-decision making
				Types of families
				The principles of the family meeting in the ICU
				Using 'VALUE' in family communications
5	9	Patient and family-centred care		Addressing conflicts within families
				Define Patient and Family-Centred Care (PFCC)
				Learn the four critical components of PFCC
				Describe strategies for delivering effective PFCC in palliative care
	10	Assessment and Management of Pain		Understand and reduce barriers to the delivery of PFCC
				Pathophysiology and types of pain
				Need for pain assessment in the ICU
				Assessment of pain in the ICU Subjective self-report assessment tools Objective behavioural assessment tools Pain Management in the ICU

				Non-pharmacological management Pharmacological management
6	11	Assessment and Management of Dyspnoea		Pathophysiology of dyspnoea in the terminally ill
				Quantification of dyspnoea Intensity Severity Impact
				Management of dyspnoea in the ICU Pharmacological management – opioids, anxiolytics, oxygen therapy Non-pharmacological management
	12	Assessment and Management of Respiratory Secretions/Thirst/Xerostomia		Evidence-based recommendations for airway hygiene in the ICU
				Assessment of respiratory secretions
				Management of respiratory secretions Pharmacological management Non-pharmacological management
				Management of thirst and xerostomia
7	13	Assessment and Management of Delirium/Agitation/Restlessness		Causes and types of delirium in ICU
				Assessment of delirium
				Management of delirium Pharmacological management Non-pharmacological management
	14	Management of psychospiritual distress in the ICU		Describe screening for psychospiritual pain in the ICU
				Describe approaches to managing spiritual and emotional distress of patients and their caregivers
				Identify thresholds for referral to mental health clinicians
8	15	Management of Refractory Distress		Describe persistent myths about palliative sedation
				Explain the circumstances under which palliative sedation may be indicated
				Describe recommended dosing for palliative sedation
				Review the potential alternatives to palliative sedation
	16	Grief and Bereavement		Review theories of grief
				Grief in sudden death
				Assess grief in patients/family members

9				Discuss strategies to address grief in patients/families
				Principles of bereavement support to families
	17	Devices and discontinuation of devices		Overview of cardiac rhythm management devices
				Deactivation of implantable cardioverter defibrillators and pacemakers – differences and challenges
				Logistics of deactivation
				Modes of deactivation
				Ethical and legal concerns in the deactivation
				Communication issues
	18	Organ donation and body donation		WHO guiding principles on human cell, tissue, organ transplantation
				Legal and ethical aspects of organ and body donations
				Effective communication in organ donation
10	19	Discharge against medical advice		Prevalence of DAMA
				Causes and factors influencing DAMA
				Consequences of DAMA
				Ethical aspects of DAMA
				Legal aspects of DAMA
	20	Self-Care for Professionals		Understand how self-care is a core competency in palliative and end of life care
				Recognise what triggers stress and burnout
				Develop a self-care plan that reduces stress and fosters personal growth and well-being
11	21	Team Collaboration and Effectiveness		Identify the conceptual basis for teamwork in ICU palliative care
				Describe different types of teams in palliative care
				Name specific advantages and challenges of teamwork
				Principles of Intra-team and Inter-team communication in ICU-PC
	22	Introducing Quality Improvement in ICU-PC		Discuss why a quality improvement methodology is an essential approach for integrating palliative care into ICU services
				Describe the basic concepts of quality improvement approaches
				Describe the first steps to improving quality
				Complete a performance improvement project
	23	PC in Paediatric ICU		Define paediatric palliative care (PPC)
				Learn how children of different developmental capacity understand concepts of illness and death and how this can impact care planning
				Use developmentally based strategies to identify and address grief in children
				Discuss grief related to the loss of a child and strategies to address family grief

12	24	PC in Perinatal/Neonatal ICU		Discuss symptom management in children
				Define neonatal palliative care and who need it
				Principles of symptom management in neonates Pain Respiratory distress/secretions Agitation Compassionate extubation Anticipatory guidance for lactating mothers
				Addressing existential distress and psychosocial care
				Addressing bereavement in families

PROGRAM OVERVIEW

Title

“EOLC & Palliative Care ICU Course (EPIC)”

- Duration: 6 months
- Online sessions: Every 2 weeks (2 hours each)
- Structure per session:
 - Interactive lecture – 30-45 min
 - Small group discussion (breakout sessions) – 45-60 min

Module 1.

- Introduction to course - 10 min
- Small group breakout discussion (core learning) - 45 min
 - Discuss the terms palliative care, EOLC, Passive euthanasia, Euthanasia, Chronic illness, terminal illness
- Talk 1 – What is palliative care why does it matter in ICU - 30 min
- Talk 2 - What is end-of-life care and what are the different related terms - 30 min

Module 2.

- Illness Trajectories and Prognostication
- Ethical & Legal Issues in EOLC
- Small group breakout discussion (core learning) - 45 min

Module 3.

Basic communication

Complex Communication Situations

Small group breakout discussion (core learning) - 45 min

Module 4.

- Goals of care discussion and decision-making
- Handling difficult families
- Small group breakout discussion (core learning) - 45 min

Module 5.

- Patient and family-centred care
- Assessment and Management of Pain
- Small group breakout discussion (core learning) - 45 min

Module 6.

- Assessment and Management of Dyspnoea
- Assessment and Management of Respiratory Secretions/Thirst/Xerostomia
- Small group breakout discussion (core learning) - 45 min

Module 7.

- Assessment and Management of Delirium/Agitation/Restlessness
- Management of psychospiritual distress in the ICU
- Small group breakout discussion (core learning) - 45 min

Module 8.

- Management of Refractory Distress
- Grief and Bereavement
- Small group breakout discussion (core learning) - 45 min

Module 9.

- Devices and discontinuation of devices
- Organ donation and body donation
- Small group breakout discussion (core learning) - 45 min

Module 10.

- Discharge against medical advise
- Self-Care for Professionals
- Small group breakout discussion (core learning) - 45 min

Module 11.

- Team Collaboration and Effectiveness
- Introducing Quality Improvement in ICU-PC
- Small group breakout discussion (core learning) - 45 min

Module 12.

- PC in Paediatric ICU
- PC in Perinatal/Neonatal ICU
- Small group breakout discussion (core learning) - 45 min
- Final Component:
2-day in-person contact session (skills-heavy)

BREAKOUT DESIGN (THIS IS YOUR PROGRAM'S STRENGTH)

Each breakout group (6–8 participants):

- 1 facilitator
- 1 case scenario
- Structured worksheet:
 - What is happening?
 - What are your concerns?
 - What would you say/do?
 - What are the ethical/legal issues?

End with 2–3 key takeaways per group

2-DAY CONTACT SESSION (CORE SKILLS LAB)

Focus: Communication Mastery + Real ICU Scenarios

Day 1: Foundations of Communication

Modules:

- Breaking bad news (SPIKES in ICU context)
- Discussing prognosis & uncertainty
- Handling emotions (anger, denial, silence)

Methods:

- Demonstration videos
- Role play with feedback
- Faculty modeling

Day 2: Advanced Conversations

Modules:

- Goals-of-care discussions
- DNAR & withdrawal conversations

- Conflict resolution (family vs team)
- Death communication & bereavement

Methods:

- Simulated ICU scenarios
- Standardized patients (if possible)
- Video recording + feedback

ASSESSMENT STRATEGY

During Online Program:

- Participation in breakout discussions
- Reflective notes (short, optional)

During Contact Session:

- Mini-CEX (communication scenarios)
- Faculty feedback

Optional:

- Pre-post confidence/self-efficacy scale

OUTPUTS FOR PARTICIPANTS

- ICU EOLC checklist toolkit
- Sample documentation templates (India-specific)
- Communication phrase bank
- Case compendium

Competency based curriculum

Competencies linked to EOLC steps

EOLC Step	Core Competency	Learning Objectives (Learner will be able to...)	Key Topics	Skills (Observable)	Teaching– Learning Methods
1. Recognize dying patient	Clinical recognition & prognostication	Identify patients with poor prognosis and likely dying trajectory	Prognostication, medical futility, illness trajectories	Identify EOLC candidates, interpret clinical course	Bedside teaching, case-based discussions (CBD), mortality reviews
2. Initiate EOLC discussion	Communication initiation	Initiate timely and appropriate EOLC conversations	Breaking bad news, uncertainty communication	Start goals-of-care discussion, use SPIKES	Role play, simulation, observed encounters
3. Assess	Respect for	Elicit	Advance care	Identify	Bedside

preferences & values	autonomy & cultural competence	patient/family values and preferences	planning, surrogate decision-making, cultural context	surrogate, explore values, document preferences	interviews, reflective exercises, small group discussions
4. Conduct family meeting	Family-centered communication	Conduct structured multidisciplinary family meetings	Family dynamics, conflict resolution, empathy	Lead family meeting, manage emotions, summarize plan	Simulation, video-based learning, Mini-CEX
5. Achieve consensus on goals of care	Shared decision-making & ethics	Facilitate consensus aligning medical and family perspectives	Ethical principles (autonomy, beneficence), shared decision-making	Negotiate goals, resolve disagreement, document consensus	Ethics case discussions, facilitated group dialogue
6. Decide withholding/withdrawal	Ethical & clinical decision-making	Apply ethical reasoning to limit life-sustaining treatment	Withholding vs withdrawal, futility, proportionality	Recommend limitation of treatment, justify decisions	Case-based ethics teaching, OSCE scenarios
7. Legal documentation	Legal literacy & compliance	Document EOLC decisions as per Indian legal framework	DNAR, Supreme Court guidelines, PMB/SMB process	Complete forms, maintain documentation, ensure consent	Workshops, document audits, supervised practice
8. Implement EOLC plan	Systems-based practice & teamwork	Translate decisions into actionable ICU care plans	ICU protocols, deprescribing, team coordination	Write EOLC orders, coordinate with nursing team	Bedside supervision, protocol-based teaching
9. Symptom control & comfort care	Clinical palliative care	Manage end-of-life symptoms effectively	Pain, dyspnea, delirium, palliative sedation	Prescribe opioids/sedatives, titrate comfort measures	Bedside teaching, skill stations, case discussions
10. End-of-life support & after-death care	Compassionate care & bereavement support	Provide dignified death and support family post-death	Death communication, bereavement, rituals	Declare death sensitively, support family, conduct debrief	Role play, reflective practice, debrief sessions

Cross-Cutting Competencies

Competency	Learning Objectives	Teaching Methods
Communication mastery	Demonstrate empathy, clarity, and structured communication	Simulation, role play, Mini-CEX
Ethical reasoning	Apply ethical principles in ICU decisions	Ethics rounds, case discussions
Legal awareness (India)	Practice within Indian legal framework	Workshops, policy teaching
Cultural & spiritual sensitivity	Respect diverse beliefs and practices	Narrative medicine, reflective writing
Teamwork & leadership	Lead interdisciplinary care	Team-based simulation
Professionalism & self-care	Manage moral distress and burnout	Reflective sessions, peer discussion

Annexure 3

SJMC Workshops –

Qualitative research workshop

A qualitative research methods training workshop series was conducted over a three-week period on 28 February, 7 March, and 14 March by the Pain and Palliative department, SJMCH. The workshops were facilitated by experts in qualitative research methodology and focused on strengthening research competencies relevant to the project. Domain experts included those from the St. John's Research Institute (SJRI), Bengaluru; the National Institute of Mental Health & Neurosciences (NIMHANS), Bengaluru; and the University of Pittsburgh, USA. A total of 25 participants were included, comprising clinicians, residents, researchers from SJMCH and SJRI, researchers from NIMHANS, and researchers from the Swami Vivekananda Youth Movement (SVYM). The workshop covered lecture, discussion and hands-on training on

- Orientation to qualitative research paradigm
- Where do we fit in the qualitative methods in critical care, and what approaches to choose from?
- Crafting a qualitative research protocol
- Connecting the dots: Insights from implementing Qualitative research in Critical Care
- Voices from the field: data collection (Interviews and focus group discussions)
- From field to findings: rigour and ethics in qualitative inquiry
- Thematic analysis in action (Coding, naming and defining themes, category development)
- Digital tools for qualitative research (Introduction to the digital tools and hands-on training on NVivo coding)

Implementation research workshop

Objectives

When participants return to their ICUs after the workshop, what are we hoping they will do?

1. A shared understanding of the evidence and policies for palliative care in the ICU
2. Reflect on the implementation gaps and strategies to address it.
3. Teams/participants go back with a problem statement/theme they can work on.

Including team/stakeholders, measurement & project/study design.

Note: Room layout - movable chairs and tables for group work. Avoid auditoriums with fixed seats.

Attendees: Ideally 3-4 people from the same unit, including nurses, intensivists, MDT palliative care personnel, and others who are involved.

Agenda

Time	Topic	Session Format	Speaker
Day 1, Friday, June 26			
2:00 - 2:30 PM	Welcome and housekeeping		Dr Vallath – St Johns Dr Seth – Univ Pittsburgh
	Intros, motivations, teams	Speed Networking - 3 rounds - 3 mins each. Conversation points	

		- What aspect of life makes today worthwhile? - What's something you did recently, at work or outside work, that left you feeling happy or satisfied & what is one <i>specific</i> thing you wish you could magically improve for patients in your unit?	
2:30 - 2:45 PM	ICMR opportunities and current work in Implementation science	Plenary – Online	Dr Neha Dahiya, Dr Ashoo Grover (ICMR)
2:45 - 3:00 PM	How has policy and implementation science worked for the greater good - lessons from prior successes (& failures)	Plenary - Online	Dr Swarup Sarkar, former WHO SEARO Director
10 min break	Tea		
3:15 - 3:35 PM	Improvement basics, examples & how it fits with implementation science Guidance for group work	Plenary - guidance for group work. teams & facilitators assigned.	Dr Vaid --WHOCC, Delhi
3:35PM - 4:45 PM	Identifying the problem - teams make a list of opportunities for improvement and rank by impact on patient outcomes, feasibility, resources, acceptability etc.	Team Studio Group Work 1	Dr P Chatterjee Dr Vaid-WHOCC, Delhi Dr A Chatterjee Dr Bhavna / Dr Vaid

	(Assign sub-themes or they pick what they like?)		
4:45 - 5 PM	Report back	One thing that you heard that you'd like to share – max 3-4 people	Bhavna, N Vallath
Day 2, Saturday, June 27			
12:00 - 12:15 PM	Recap prior day, Qs	Speed Networking - 3 rounds - Something new I learnt yesterday - Something that I connect strongly with - Something I don't understand (other person tries to help)	Dr Pratyasa, Dr Vaid
12:15 - 12:45 PM	CFIR framework: Understanding systems and barriers & facilitators	Plenary Followed by 15 min Q & A	Drs Parvathy/Garima
12:45 - 1:30 PM	Lunch break		
1:30 -2:30 PM	Expert panel, current policy change, foreseeable facilitators & barriers. Policy/upstream factors as barriers/facilitators to pall care adoption, how research may help +- solve	Panel moderated by Dr N Salins	ICU/ Pall care experts Dr Bhuvana, Dr Anuja, Dr Rajesh Shetty, Assigned Intensivists, CMS, ANS critical care
2:40 PM - 3 PM	Outcomes and measurement, phases: adoption, fidelity, feasibility, acceptability, appropriateness, penetration,	Plenary	Dr Prem Mony, SJRI

	sustainability		
2:30 - 2:40 PM	Break 10 min		
3:00 - 4:00 pm	Applying CFIR to an implementation question/ project theme, based on what was discussed	Team Studio Group Work 2 (Posters on wall with specific themes, people go work on the theme assigned List strategies on slide	Facilitators, Moderators Dr Bhuvana/Dr Vallath/Dr Mony/Assigned intensivists/ Garima/ Dr Arun
4PM - 5PM	Pall care Impl Sci US	Plenary	Dr Gina Piscitello, University of Pittsburgh, (online)
Day 3, Sunday, June 28			
9 AM - 9:30AM	Study example in use - colon cancer screening adoption RCT	A study applying the principles learnt the previous day	Dr Divya Shankar-Boston University(online)
10:15 AM - 10:30 AM	Implementation strategies: ERIC taxonomy, tailoring, and strategy specification	Plenary	Dr Parvathy/Dr Pratyasha
10:30 - 11 AM	Discuss possible measures/ available data for integrating palliative care in ICU. What is available? What is not.	Team Studio Group Work 3 (using project ideas/themes list available data)	
11 AM - 11:20 AM	Implications of policy research on implementation of operationalisation of evidence - especially in the Indian context	Plenary	Dr Anirban Chatterjee - University of South Caroline
11:20AM - 11:30AM	What makes an implementation idea publishable	Plenary	Dr Nagi – Publishing - Springer Nature Journal
10 minutes	Break		
11:40 - 12:30	Team Studio : top determinants,	Team Studio Group Work 4 - interactive format -	Dr Vaid, former WHO

	strategies, outcomes and priorities	TRIZ 1) What we need to 'stop' doing 2) Commitments we can make	
12:30 - 1PM	Feedback, suggestions, Validictory, Certificate announcement		
Lunch			